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Introduction

For our final project, the dataset we decided to explore was the Insurance dataset. We focused on the main question: Can we successfully predict insurance charges? To gain a better understanding on our main objective, we specifically investigated three key sub-questions:

- Do older individuals tend to have higher insurance charges?
- Does smoker status have a significant impact on insurance charges?
- Do smokers or non-smokers pay more insurance charges depending on their BMI?

Our motivation for this research comes from the concern that insurance costs are very expensive these days. We want to investigate the specific factors leading to an increase in insurance costs. Our goal is to advocate for a healthcare system that is both accessible and affordable for all.

Data Description

For the data description, we looked at the independent variables: 'Age,' 'Sex,' 'BMI,' 'Children,' 'Smoker,' and 'Region.' These include continuous quantitative variables such as 'Age,' 'BMI,' and 'Charges,' as well as categorical variables like 'Sex,' 'Children,' 'Smoker,' and 'Region.' The dependent variable, 'Charges,' are the insurance charges in USD.

The summary statistics of the dataset provide insights into the central tendency and spread of each quantitative variable. For instance, the mean insurance charge is \$13,270, ranging from a minimum of \$1,122 to a maximum of \$63,770.

Upon examining scatter plots of relationship between the response variable and each independent variable, we noticed interesting associations. For example, the scatter plots for 'Smoker,' 'Region,' 'Sex,' and 'Children' suggest that a bar plot may better represent the data since they are all categorical. Furthermore, the scatter plots for 'Age' and 'BMI' indicate a very weak to no association with 'Charges'.

Exploratory Data Analysis

Table 1: Dataset

age	sex	bmi	children	smoker	region	charges
19	female	27.900	0	yes	southwest	16884.924
18	male	33.770	1	no	southeast	1725.552
28	male	33.000	3	no	southeast	4449.462
33	male	22.705	0	no	northwest	21984.471
32	male	28.880	0	no	northwest	3866.855
31	female	25.740	0	no	southeast	3756.622

Dependent Variable

Moving on to the Exploratory Data Analysis (EDA), we notice that the histogram for the 'charges' variable in figure 1 appears to have a right-skewed distribution. The data's distribution is not symmetric, with most of the data points on the left-hand side, showing that the majority of observations are smaller, while a few larger observations contribute to the right tail.

To decrease the right-skewness of the data, we applied a log transformation to the 'charges' variable. This transformation helps spread out the values, making the distribution more symmetrical.

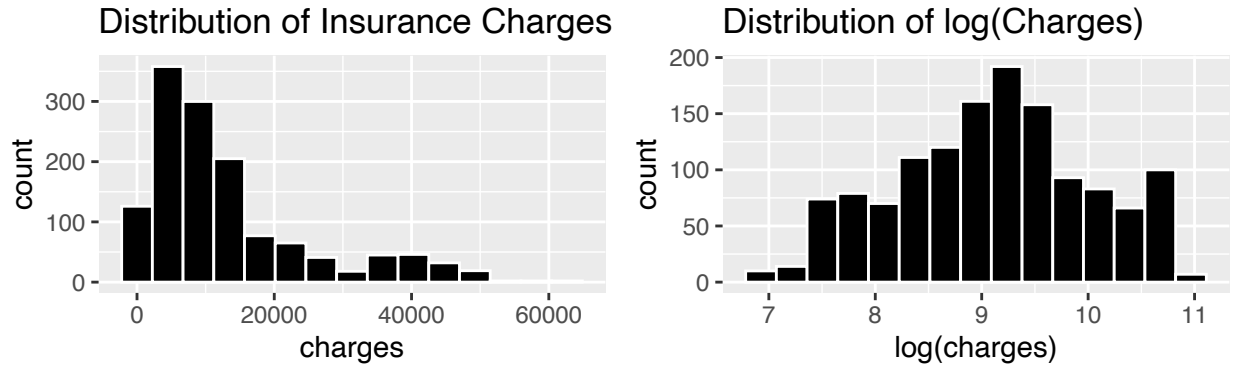


Figure 1: Charges Bar Plot

Independent Variables

When examining the graphs of the predictors in figure 2, we can see that the age histogram follows a uniform distribution with a few exceptions. This suggests that the dataset contains individuals across various age groups. The histogram for the number of children reveals a right-skewed distribution, indicating that a majority of individuals have fewer children. As you can see, BMI displays a relatively normal distribution.

Secondly, in the distribution of sexes one can see similar frequencies, which is integral to preventing bias in the predictive model. Similarly, consistent frequencies across different regions display a balanced representation of the geographical areas. Additionally, the dataset appears to consist mainly of non-smokers, indicating a class imbalance. It is important to point out that we did not find sex and region to be statistically useful so it is not included in our final model.

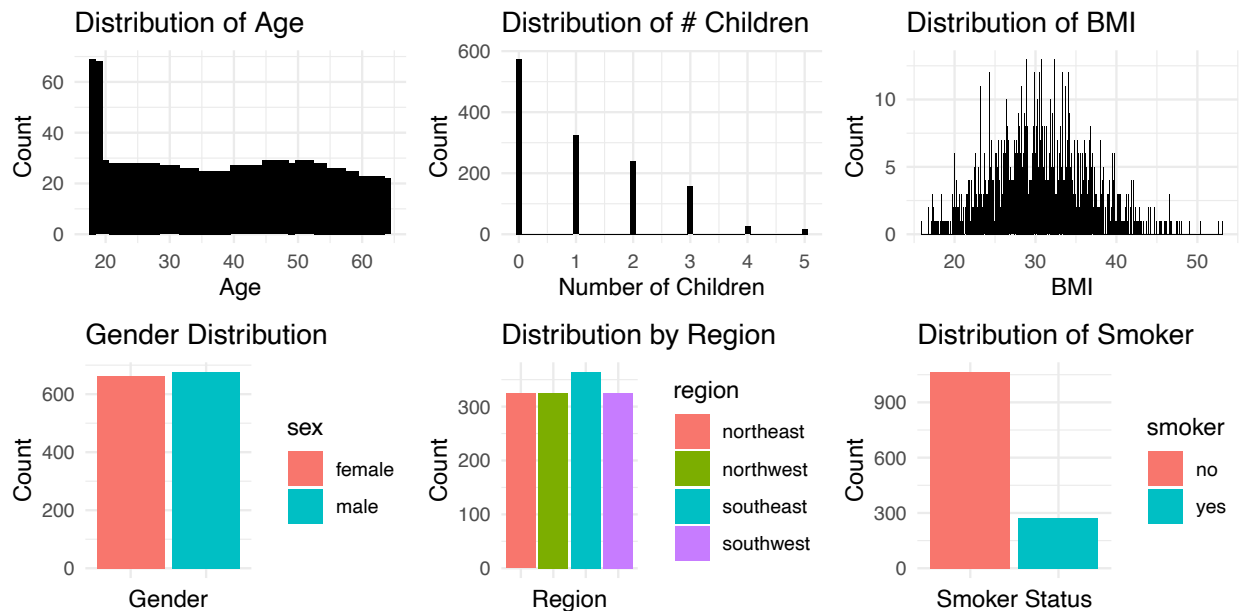


Figure 2: Distribution of Predictor Variables

Independent Variables vs Dependent Variable

When examining the relationship between predictor and response variables in figure 3, the age vs. charges scatter plot reveals three distinct sets of points, each following a separate positive linear pattern. The topmost group consists predominantly of smokers, the middle group is a mix of smokers and non-smokers, while the bottommost group consists only of non-smokers. Hence it may be ideal to make three different models if time permits. This suggests that there may be interaction effects between age and another variable.

The BMI vs. charges scatter plot indicates no obvious relationship between the two variables. The sex vs charges box plot appears to be skewed and suggests that there is a difference in the distribution of females and males. The region vs charges box plot also appears to be skewed with several outliers.

When examining the smoker vs. charges box plot it provides sufficient evidence that smokers seem to face higher charges compared to non-smokers. These observations illustrate the significance of interactions, variable relationships included in the dataset which will allow us to build a strong model.

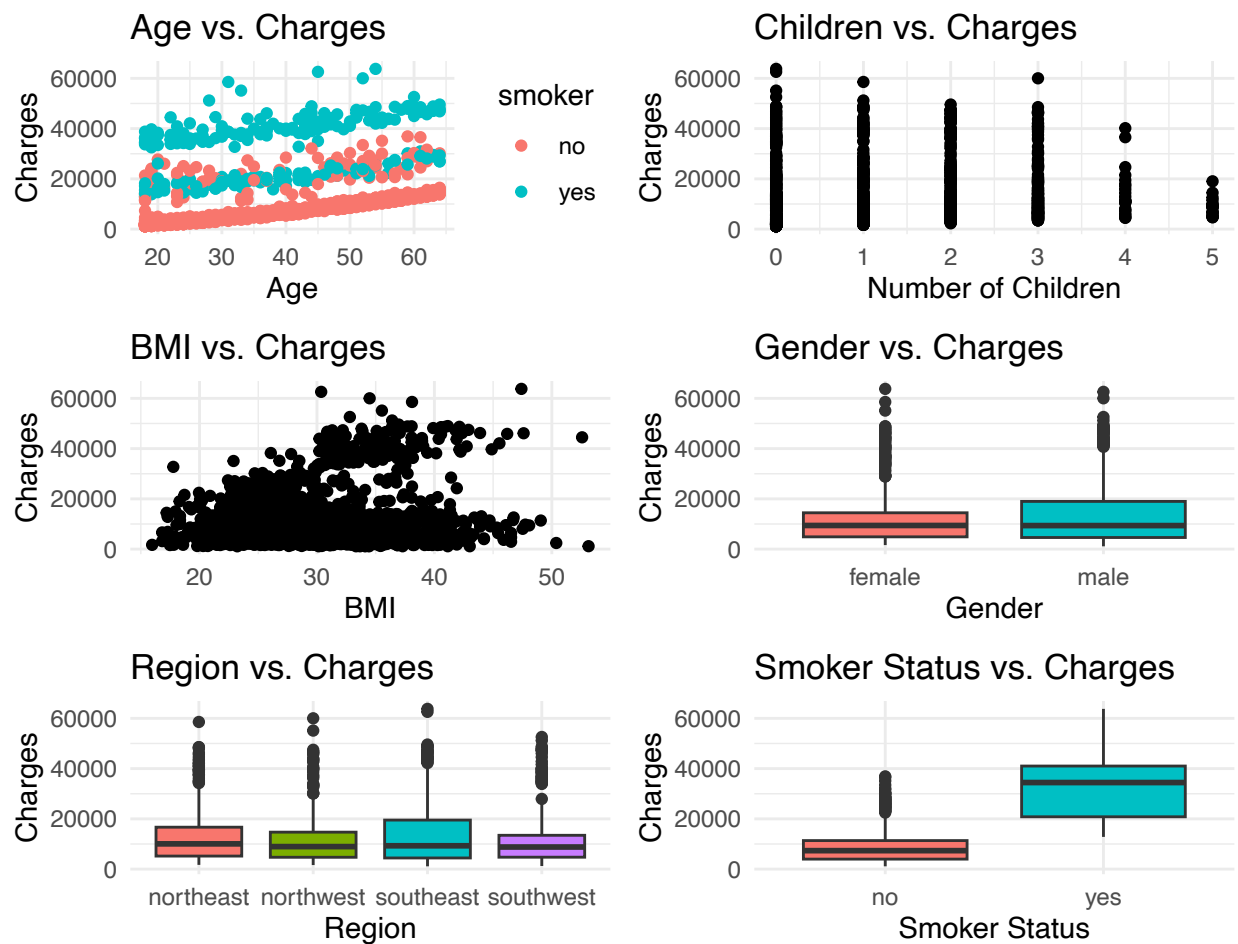


Figure 3: Predictors vs Charges

Regression Analysis

In order to find our final model we first sought to find the functional form of our base model which included all the numeric variables: age, bmi, and children. This would allow us to see if any transformations or

interactions needed to be added to the model in order to meet our assumptions. The first step we took in trying to find the functional form was by removing the contribution of variables that have a strong relationship with Y. However, this step did not improve the normality of Y, which led us to believe transformations would not be necessary for our model. The next step we took was to see if there were any interactions between variables. One interaction we found was between bmi and smoker.

This led us to add an interaction term between bmi and smoker to our model. Another addition we made to our model is by adding an intercept of 0 to the model. The reason for this is that if age was set to 0 meaning someone is not alive, it does not make sense for a dead person to incur insurance charges.

With these additions to the model we saw that the normality and linearity assumptions were met. The only assumption we were not able to meet was constant variance. We believe that there are hidden variables that we do not have access to that are causing this issue.

We then executed step wise regression to find the best possible model with our new base model. However, the stepwise regression algorithm removes the intercept of 0 which is crucial to our model. Due to the addition, of the y-intercept, which does not have a meaningful interpretation, the R^2 was decreased greatly. We concluded that step wise regression would not improve our model based on this.

Our final model ends up being: $\log(charges) = 0 + 0.071486(age) + 0.245349(children) + 0.229045(bmi) - 0.050251(bmi)(smoker) + \epsilon$

Coefficient for age: holding other variables constant, a one unit increase in age is associated with a 0.071486 increase in the logarithm of charges. (p-value < 2e-16) Coefficient for children: holding other variables constant, a one unit increase in children is associated with a 0.245349 increase in the logarithm of charges. (p-value 3.19e-15) Coefficient for bmi: holding other variables constant, a one unit increase in bmi is associated with a 0.229045 increase in the logarithm of charges. (p-value < 2e-16) Coefficient for interaction between being a smoker and bmi: each one unit increase of BMI is associated with a decrease of 0.050251 units in log-transformed charges, holding other variables constant. (p-value < 2e-16)

The R^2 value of our model is 0.9777. This means that 97.77% of the variability in insurance charges can be explained by our model.

The residual standard error of our model is 1.366 on 1334 degrees of freedom. Our RSE is pretty close to 1 which is why we feel that our model is quite accurate in predicting in insurance charges.

Conclusion

Model Overview

Despite our efforts to achieve constant variance through the creation of several different models, we encountered challenges in satisfying both assumptions. Despite this, our final model showcases a high R-squared value of 0.9777, indicating that a major portion of the variability in the logarithm of charges can be explained by the selected predictors. In addition, 3/4 of the assumptions seemed to be satisfied by our current model.

Final model:

$$\log(charges) = 0 + \beta_1 * age + \beta_2 * children + \beta_3 * bmi + \beta_{4,5} * bmi : smoker + \epsilon$$

where:

- $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients for the selected predictors
- ϵ represents the error term

The F-statistic of 1.465e+04 and the low p-value (< 2.2e-16), provide strong evidence for the overall significance of our model in accurately predicting the logarithm of charges. While we weren't able to satisfy the constant variance assumption, the high metrics show the model's ability to provide valuable insights regarding the relationship between the selected predictors and transformed dependent variable.

Interpretation

Table 2: Summary of Model Coefficients

Variable	Estimate	Std..Error	t.value	Pr...t..
age	0.071486	0.002464	29.014	< 2e-16 ***
children	0.245349	0.030755	7.978	3.19e-15 ***
bmi	0.229045	0.004036	56.753	< 2e-16 ***
bmi:smoker	-0.050251	0.002957	-16.996	< 2e-16 ***

Table 3: Additional Model Statistics

RSE	DF	Multiple.Rsquare	Adjusted.Rsquare	F.statistic	F.statistic.DF	p.value
1.366	1334	0.9777	0.9777	1.465e+04	4 and 1334	< 2.2e-16

The positive coefficient for age suggests that, holding other variables constant, a one-unit increase in age is associated with a 0.071486 increase in the logarithm of charges. This aligns with the expectation that older individuals tend to experience higher insurance expenses due to increased health risks and more costly treatments. The positive coefficients for both children and bmi suggest that individuals with more children or higher BMI tend to have higher insurance charges.

The negative coefficient for bmi:smoker ($\beta_{bmi:smoker} = 0.050251$) suggests that the effect of BMI on charges is affected by smoking status. Specifically, for smokers, the relationship between BMI and charges differs compared to non-smokers. While BMI still positively influences charges for smokers, the effect is less obvious

The residuals provide insight into the model's performance, indicating the differences between observed and predicted values. The model overall demonstrates high significance, as indicated by the low p-values for the coefficients.

Multicollinearity

When running VIF we found that the values were all under 10. We removed the BMI factor since it was not outputting a VIF value since the addition of the BMI was causing there to be a almost perfect multicollinearity in our model.

Residual Diagnostics

Table 4: Shapiro-Wilk Results

	Test.Statistic	P.value
W	0.998352	0.2231491

The QQ plot and the results from the Shapiro-Wilk normality tests suggest that the residuals of the model follow a approximate normal distribution. This is good, as it implies that the assumption of normality for the model residuals is satisfied.

However, an interesting observation can be seen from the residual vs. fitted plot, where a negative slope is evident. This may be a sign for the presence of heteroscedasticity. We attempted many different transformation and addition of other independent variables into the model, however we were not able to find a

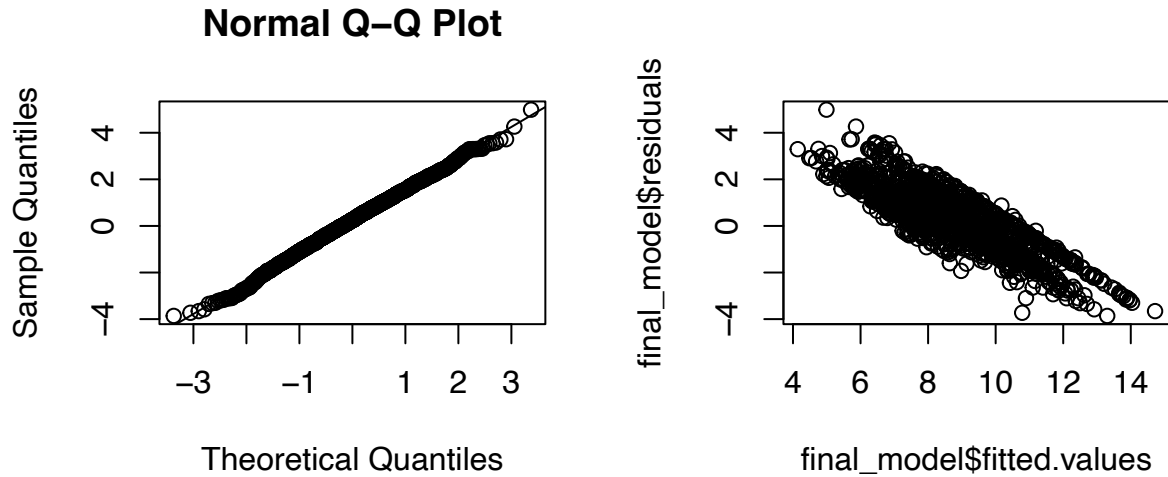


Figure 4: Assumptions

proper solution to the negative trend in the plot. The heteroscedasticity in the model implies that there are unaccounted factors that impact the variability of the residuals.

On the other hand, the assumption of independence, tested through the Durbin-Watson test, yields a p-value of 0.82. Given this high p-value, we accept the null hypothesis, indicating that the residuals show independence.

For the linearity assumption, it seems that the model is linear but has a negative slope.

In summary, while the model residuals does pass normality, linearity, and independence, the observed negative slope in the residual vs. fitted plot hints at potential heteroscedasticity. Despite multiple attempts to fix this issue, we were not able to find a proper solution.

Limitations

After trying different transformation methods, we were not able to fix the negative slope in the residuals vs fitted plot. This indicates that there might be some missing variables or interactions that cause of the downward effect in our plot.

A solution to this is to add more relevant variables or identify more interactions. Some potential variables that we thought may be helpful are credit history or health history since these are valid predictors that can help predict charges.

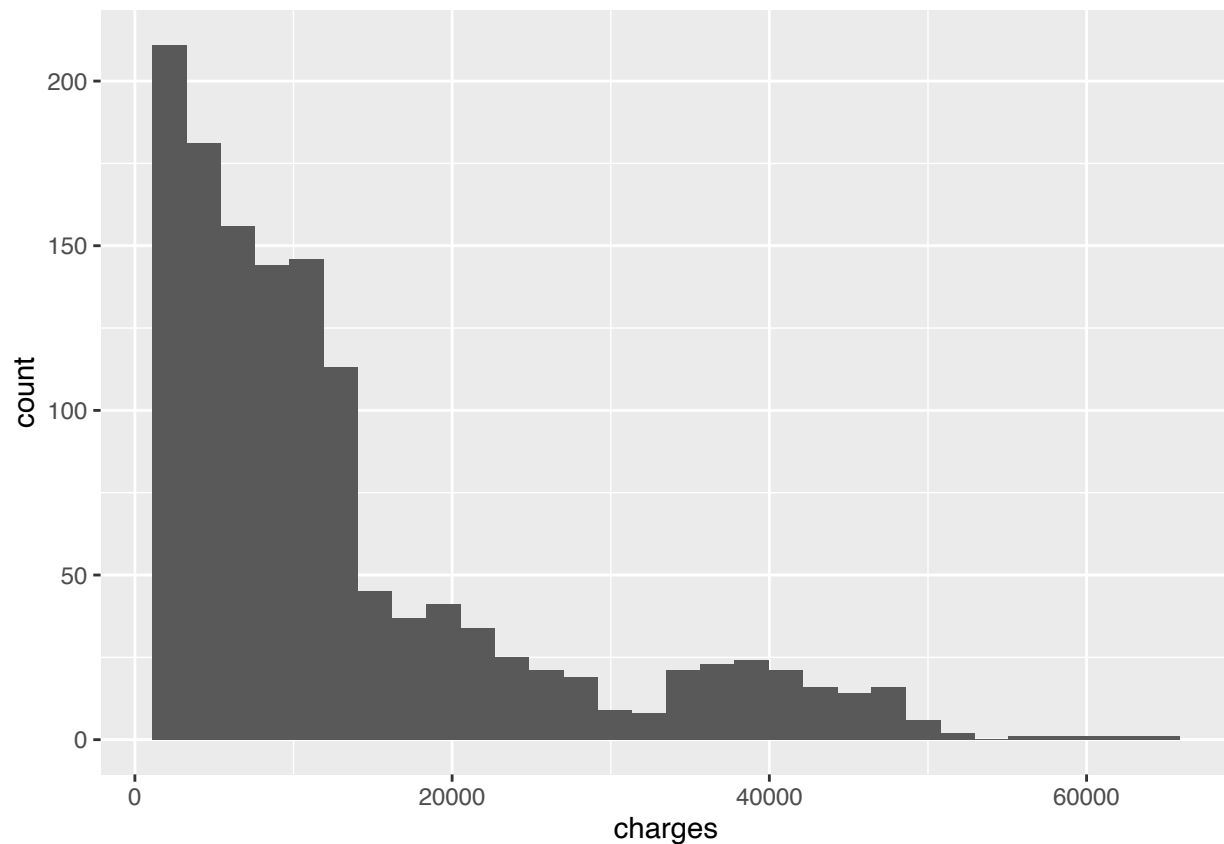
Another way to improve our model is to create three separate models. We can see from the age versus charges plot that there seems to be three different subsets so we can create individual models for each subset which may help with the negative pattern that we observe in the model. Next, having more comprehensive data will also help since we saw that in our current data set there is data missing.

Overall, we were able to take a look at potential factors that are increasing the charges for healthcare. We would like to further explore factors that contribute to this increased insurance charges and further improve our model.

APPENDIX

EDA: Response Variable

```
library(ggplot2)
ggplot(insurance, aes(x=charges)) + geom_histogram(bins = 30)
```



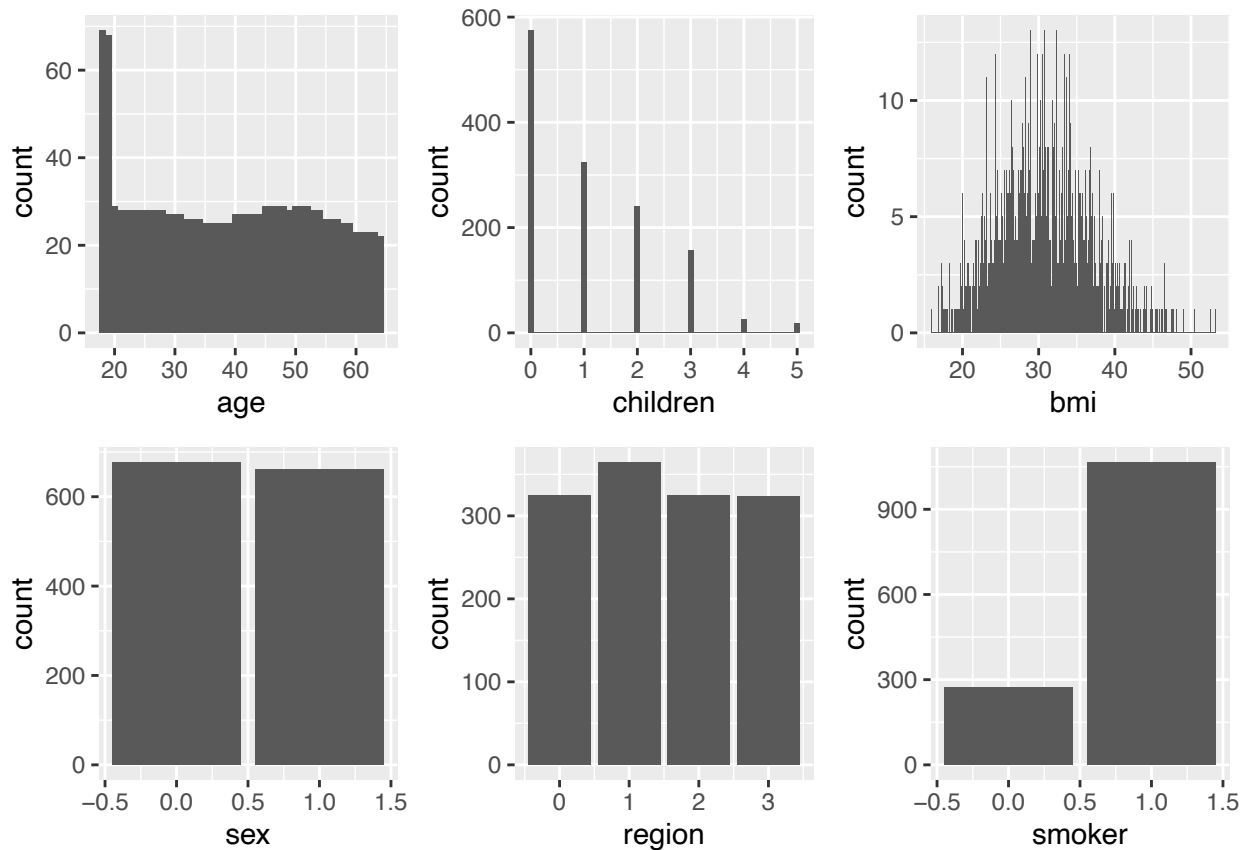
The histogram for the charges variable appears to be right-skewed. The distribution of the data is not symmetric, and the majority of the data points are concentrated on the left hand side. This suggests that most observations are smaller with a few larger observations.

EDA: Predictor Variables

```
library("ggplot2")
library("gridExtra")

# three plots and store variables
h1<-ggplot(insurance, aes(x=age)) + geom_histogram(binwidth=1)
h2<-ggplot(insurance, aes(x=children)) + geom_histogram(binwidth=0.1)
h3<-ggplot(insurance, aes(x=bmi)) + geom_histogram(binwidth=0.05)
b1<-ggplot(data=insurance, aes(x=sex))+geom_bar()
b2<-ggplot(data=insurance, aes(x=region)) + geom_bar()
b3<-ggplot(data=insurance, aes(x=smoker)) + geom_bar()
```

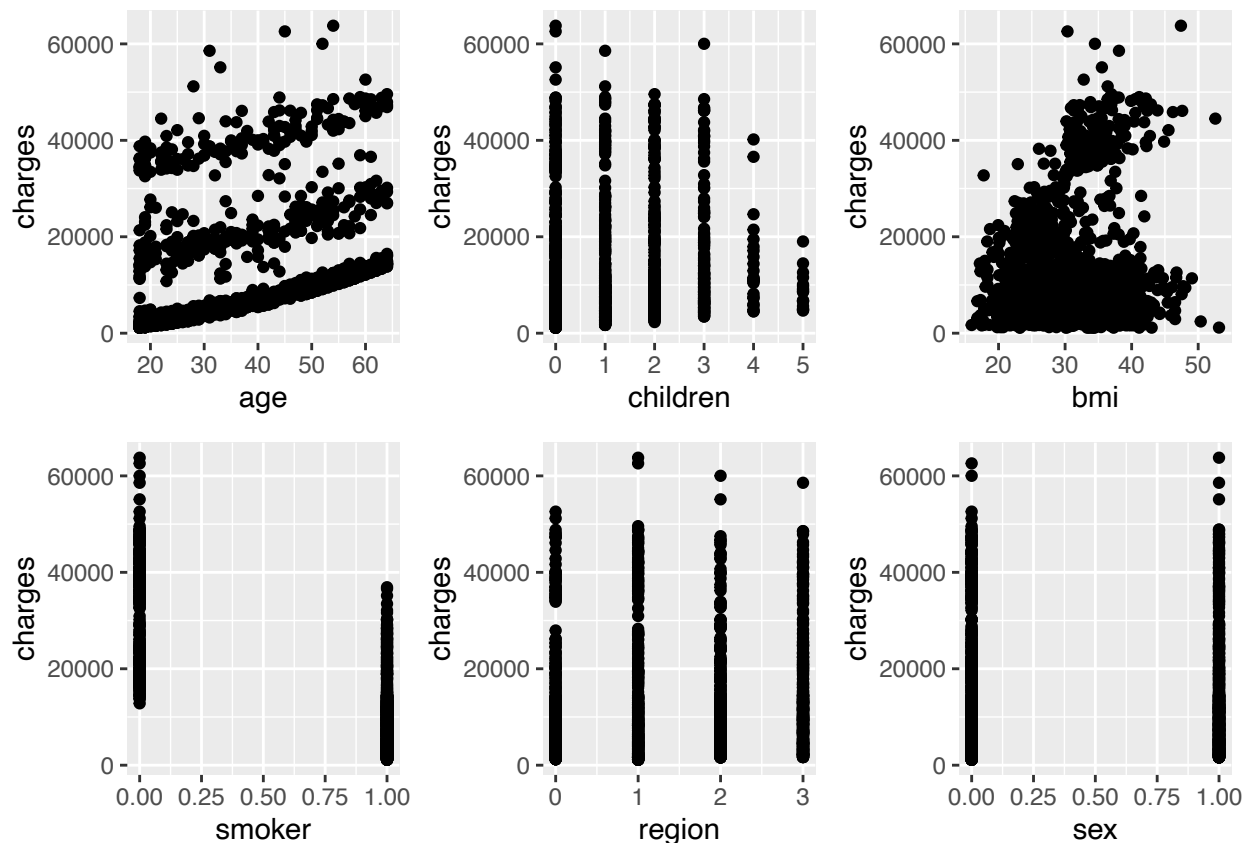
```
grid.arrange(h1, h2,h3,b1,b2,b3, ncol = 3, nrow = 2)
```



We can see that the age histogram follows a uniform distribution with a few exceptions. The histogram for the number of children is right-skewed. BMI appears to have a fairly normal distribution. From the bar graphs, we can tell that both sexes have similar frequencies, and that all regions have similar frequencies. We can also see that the majority of the data consists of non smokers.

EDA: Response Vs. Predictors

```
library(ggplot2)
library(gridExtra)
# Scatter plots of response variable with each independent variables
s1 <- ggplot(insurance, aes(x = age, y = charges)) + geom_point()
s2 <- ggplot(insurance, aes(x = children, y = charges)) + geom_point()
s3 <- ggplot(insurance, aes(x = bmi, y = charges)) + geom_point()
s4 <- ggplot(insurance, aes(x = smoker, y = charges)) + geom_point()
s5 <- ggplot(insurance, aes(x = region, y = charges)) + geom_point()
s6 <- ggplot(insurance, aes(x = sex, y = charges)) + geom_point()
grid.arrange(s1, s2, s3, s4, s5, s6, nrow = 2, ncol = 3)
```



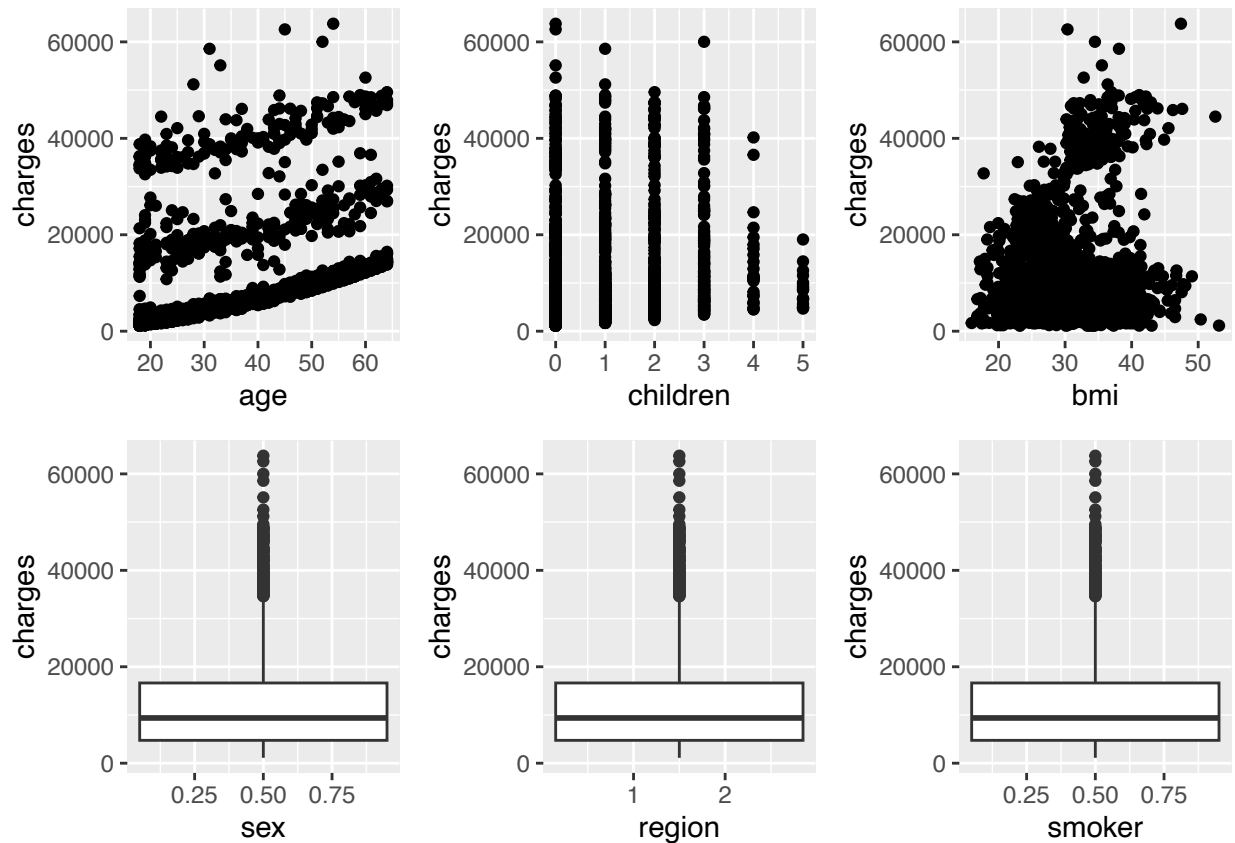
```
library("ggplot2")
library("gridExtra")

#three plots and store variables
s1<-ggplot(insurance, aes(x=age, y=charges)) + geom_point()
s2<-ggplot(insurance, aes(x=children, y=charges)) + geom_point()
s3<-ggplot(insurance, aes(x=bmi, y=charges)) + geom_point()

box1<-ggplot(insurance, aes(x=sex, y=charges)) +
  geom_boxplot()
box2<-ggplot(insurance, aes(x=region, y=charges)) +
  geom_boxplot()
box3<-ggplot(insurance, aes(x=smoker, y=charges)) +
  geom_boxplot()

grid.arrange(s1, s2,s3,box1, box2, box3, ncol = 3, nrow = 2)
```

```
## Warning: Continuous x aesthetic
## i did you forget 'aes(group = ...)'?
## Continuous x aesthetic
## i did you forget 'aes(group = ...)'?
## Continuous x aesthetic
## i did you forget 'aes(group = ...)'?
```

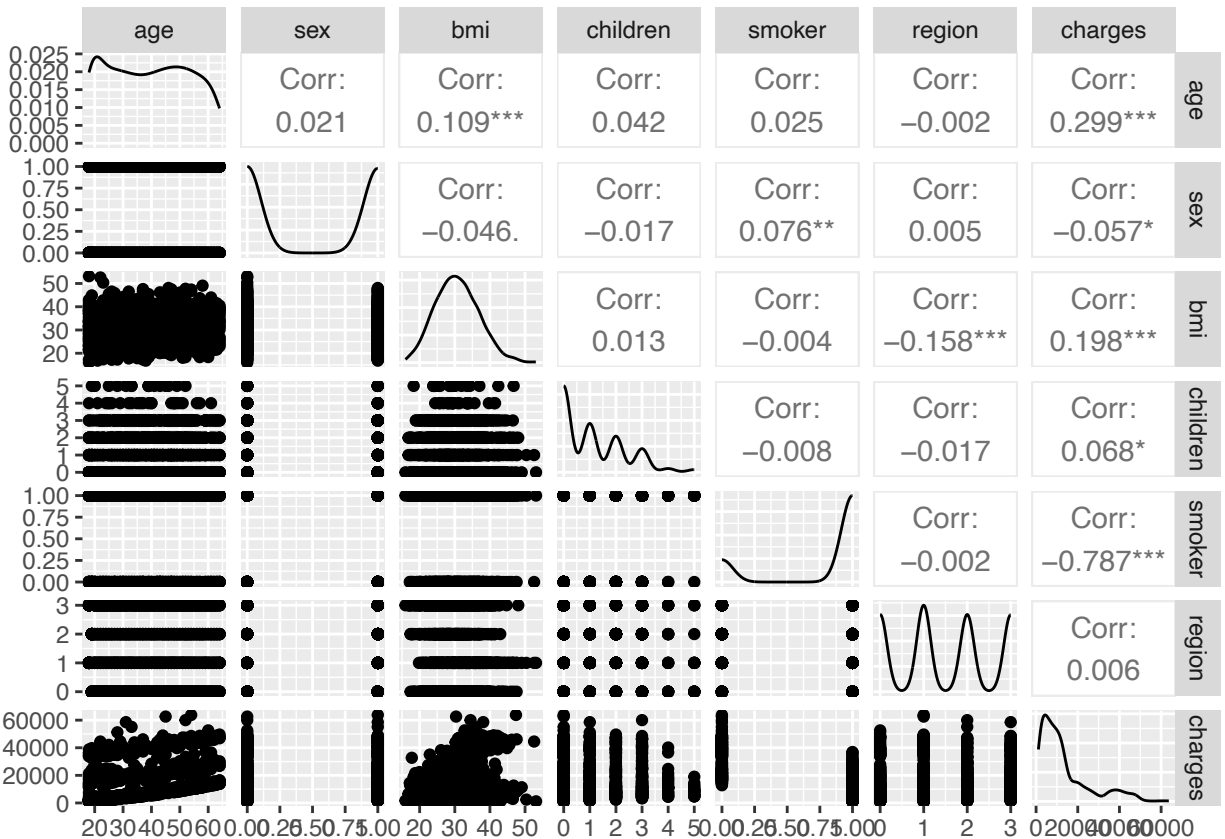


Multicollinearity analysis

```
library("GGally")
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2
```

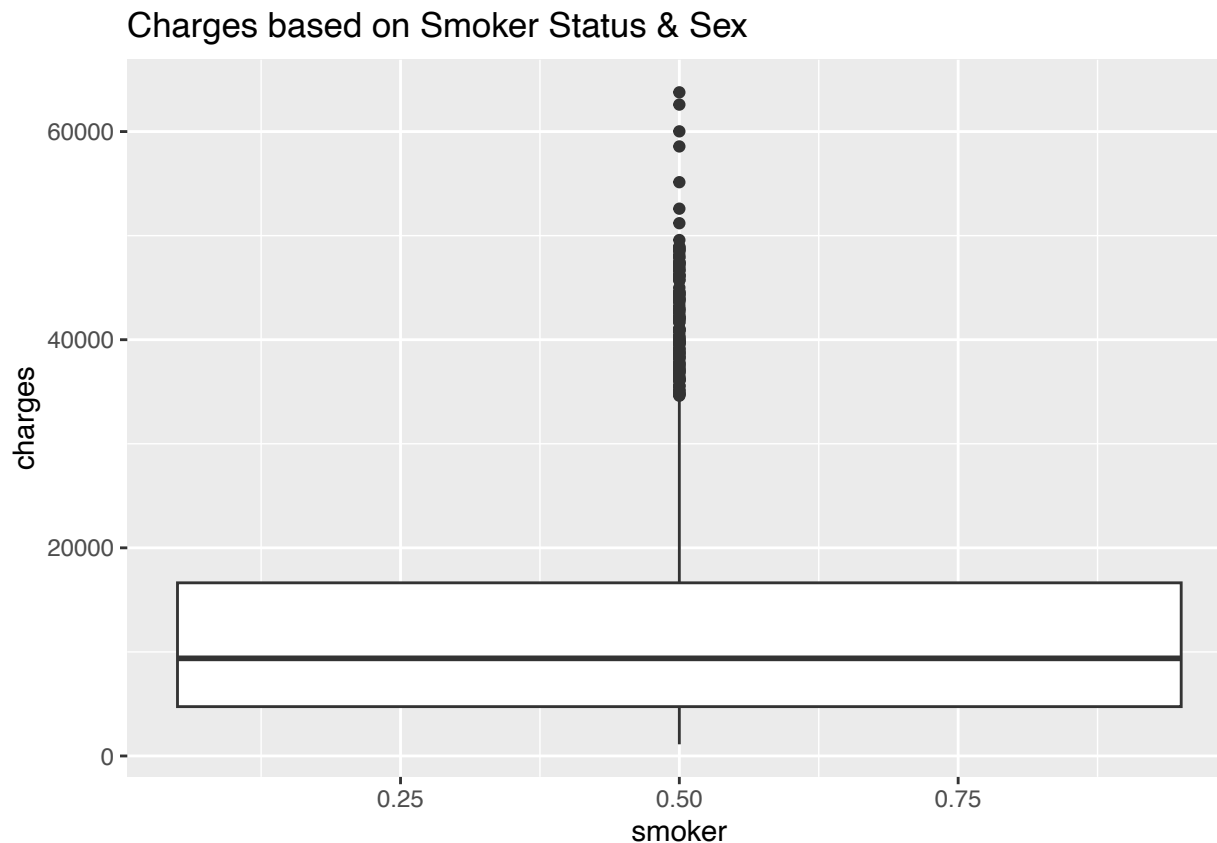
```
# pairwise scatterplots and coefficients of correlation
ggpairs(insurance) + theme()
```



```
# charges by smoker status and sex using boxplots
ggplot(insurance, aes(x = smoker, y = charges, fill = sex)) +
  geom_boxplot() + ggtitle("Charges based on Smoker Status & Sex")
```

```
## Warning: Continuous x aesthetic
## i did you forget 'aes(group = ...)'?
```

```
## Warning: The following aesthetics were dropped during statistical transformation: fill
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



Multicollinearity: Nonsignificant t-tests for all (or nearly all) independent variables when the overall F-test is significant

```
model <- lm(charges~children+age+bmi+sex+region+smoker,data=insurance)
summary(model)
```

```
##
## Call:
## lm(formula = charges ~ children + age + bmi + sex + region +
##     smoker, data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11343  -2807  -1017   1408   29752
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10812.95    1066.09  10.143 < 2e-16 ***
## children      479.37     137.64   3.483 0.000513 ***
## age           257.29      11.89  21.647 < 2e-16 ***
## bmi           332.57      27.72  11.997 < 2e-16 ***
## sex           131.11     332.81   0.394 0.693681
## region        353.64     151.93   2.328 0.020077 *
## smoker       -23820.43    411.84 -57.839 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 6060 on 1331 degrees of freedom
## Multiple R-squared:  0.7507, Adjusted R-squared:  0.7496
## F-statistic: 668.1 on 6 and 1331 DF,  p-value: < 2.2e-16
```

The F-test for H_0 : All the beta params are equal is highly significant ($F=500.8$, $p\text{-value} < 2.23 \times 10^{-16}$). Therefore we can reject H_0 for any alpha greater than 0.0001 and conclude that at least one of the beta parameters, is nonzero. The t-tests for sex and region, however, are nonsignificant.

Multicollinearity: Variance Inflation Factor (VIF)

```
library(car)
```

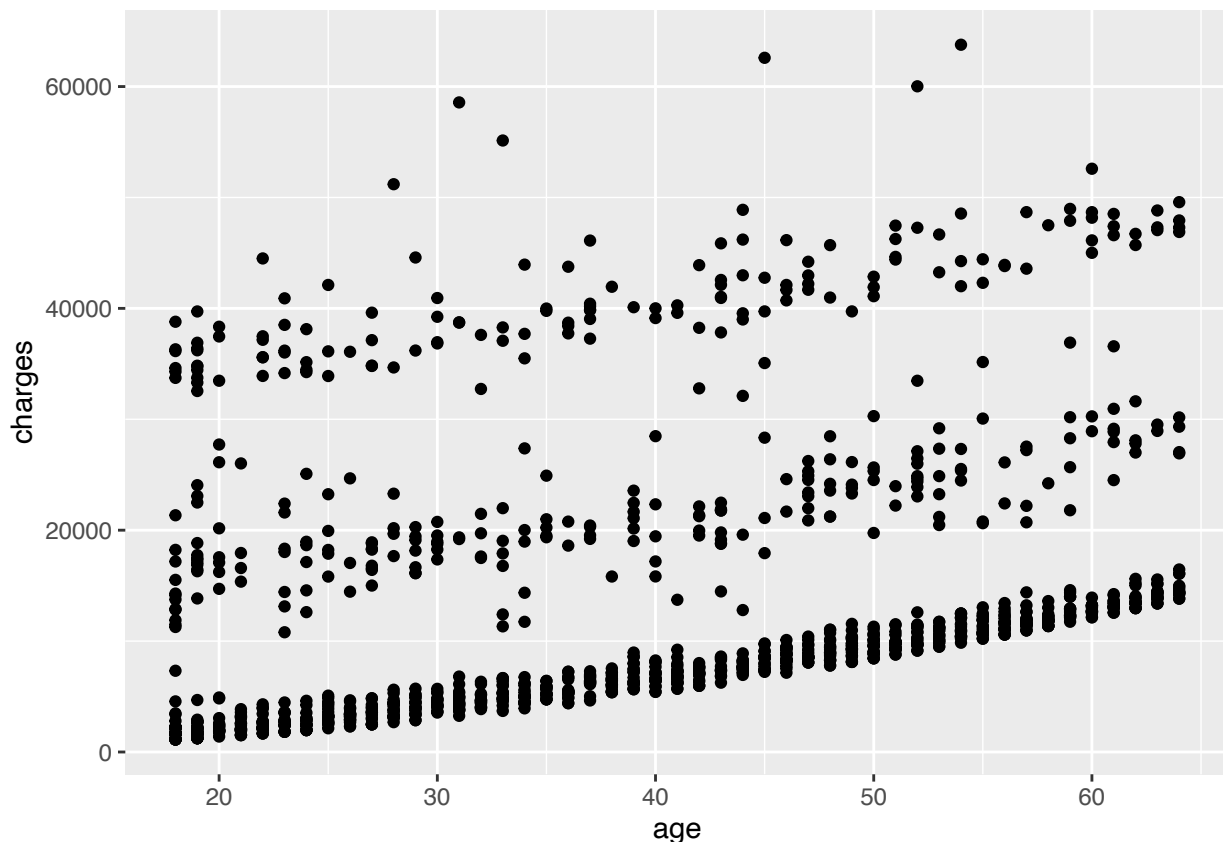
```
## Loading required package: carData
```

```
vif(model)
```

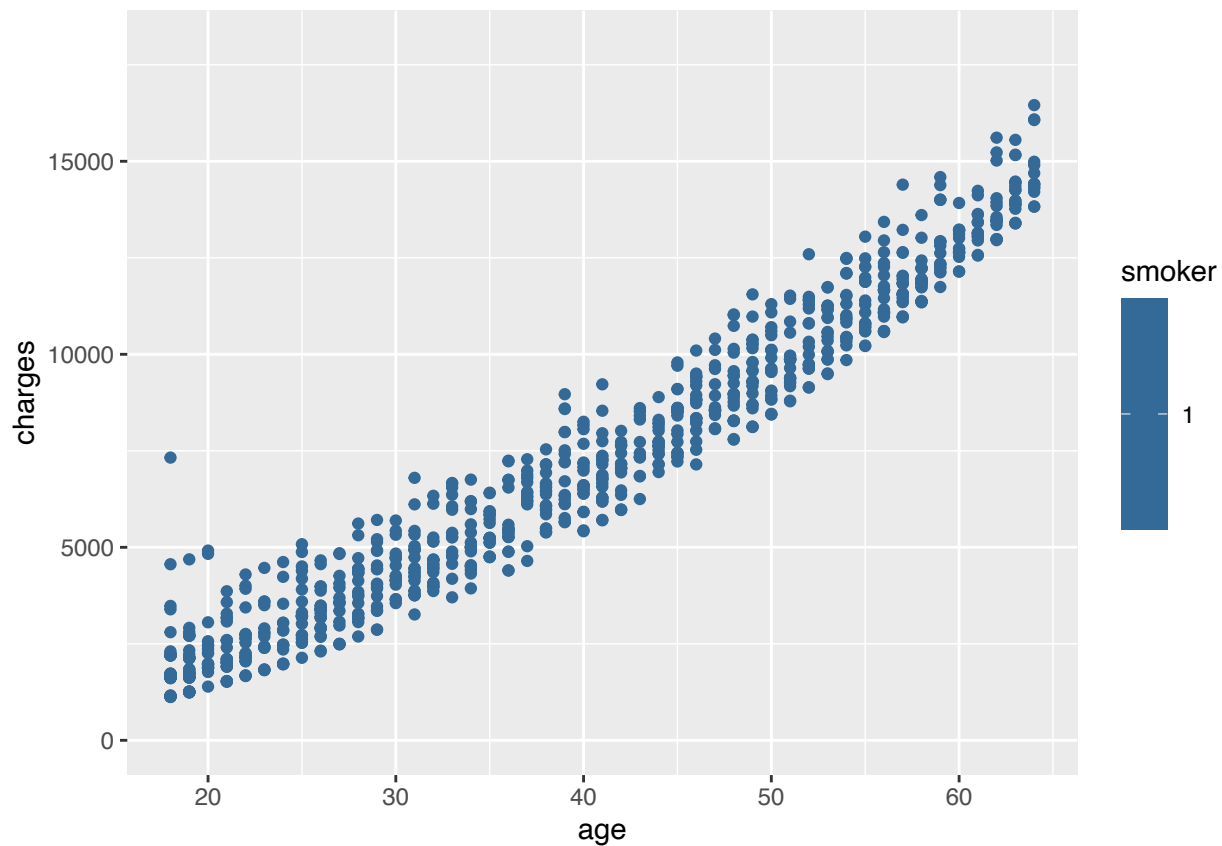
```
## children      age      bmi      sex      region      smoker
## 1.002482 1.015394 1.040608 1.008889 1.025966 1.006466
```

Since all of the VIF values are close to 1, multicollinearity isn't a problem in the model.

```
ggplot(insurance, aes(x=age, y=charges)) + geom_point()
```




```
data1 <- insurance[!(insurance$age <= 45 & insurance$charges > 10000) & insurance$charges < 18000, ]
ggplot(data1, aes(x = age, y = charges, color = smoker)) + geom_point() + ylim(c(0, 18000))
```



We noticed that there are three groups of points in the age vs charges scatter plot and decided to treat each group as a separate data set and create a model for each. For now we will be working with the group at the bottom of the set of points, which consists of 966 entries, and later replicate the steps to create models for the other two groups.

Model Selection

```
insurance$sex[insurance$sex == 'male'] <- 0
insurance$sex[insurance$sex == 'female'] <- 1
insurance$smoker[insurance$smoker == 'yes'] <- 0
insurance$smoker[insurance$smoker == 'no'] <- 1
insurance$region[insurance$region == 'southwest'] <- 0
insurance$region[insurance$region == 'southeast'] <- 1
insurance$region[insurance$region == 'northwest'] <- 2
insurance$region[insurance$region == 'northeast'] <- 3
insurance$smoker<-as.numeric(insurance$smoker)
insurance$sex<-as.numeric(insurance$sex)
insurance$region<-as.numeric(insurance$region)
head(insurance)
```

```
##   age sex    bmi children smoker region  charges
```

```
## 1 19 1 27.900      0      0      0 16884.924
## 2 18 0 33.770      1      1      1 1725.552
## 3 28 0 33.000      3      1      1 4449.462
## 4 33 0 22.705      0      1      2 21984.471
## 5 32 0 28.880      0      1      2 3866.855
## 6 31 1 25.740      0      1      1 3756.622
```

```
null_model <- lm(charges ~ 1, data = insurance)
full_model <- lm(charges ~ ., data = insurance)
#full_model <- lm(charges ~ (.)^2, data = insurance)
step_model1 <- step(null_model, scope = list(lower = null_model, upper = full_model), direction = "bot
```

```
## Start: AIC=25160.18
```

```
## charges ~ 1
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## + smoker	1	1.2152e+11	7.4554e+10	23868
## + age	1	1.7530e+10	1.7854e+11	25037
## + bmi	1	7.7134e+09	1.8836e+11	25108
## + children	1	9.0660e+08	1.9517e+11	25156
## + sex	1	6.4359e+08	1.9543e+11	25158
## <none>			1.9607e+11	25160
## + region	1	7.5571e+06	1.9607e+11	25162

```
##
```

```
## Step: AIC=23868.38
```

```
## charges ~ smoker
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## + age	1	1.9928e+10	5.4626e+10	23454
## + bmi	1	7.4856e+09	6.7069e+10	23729
## + children	1	7.5272e+08	7.3802e+10	23857
## <none>			7.4554e+10	23868
## + region	1	3.9555e+06	7.4550e+10	23870
## + sex	1	1.4213e+06	7.4553e+10	23870
## - smoker	1	1.2152e+11	1.9607e+11	25160

```
##
```

```
## Step: AIC=23454.24
```

```
## charges ~ smoker + age
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## + bmi	1	5.1129e+09	4.9513e+10	23325
## + children	1	4.5928e+08	5.4167e+10	23445
## <none>			5.4626e+10	23454
## + region	1	5.2055e+06	5.4621e+10	23456
## + sex	1	2.2255e+06	5.4624e+10	23456
## - age	1	1.9928e+10	7.4554e+10	23868
## - smoker	1	1.2392e+11	1.7854e+11	25037

```
##
```

```
## Step: AIC=23324.76
```

```
## charges ~ smoker + age + bmi
```

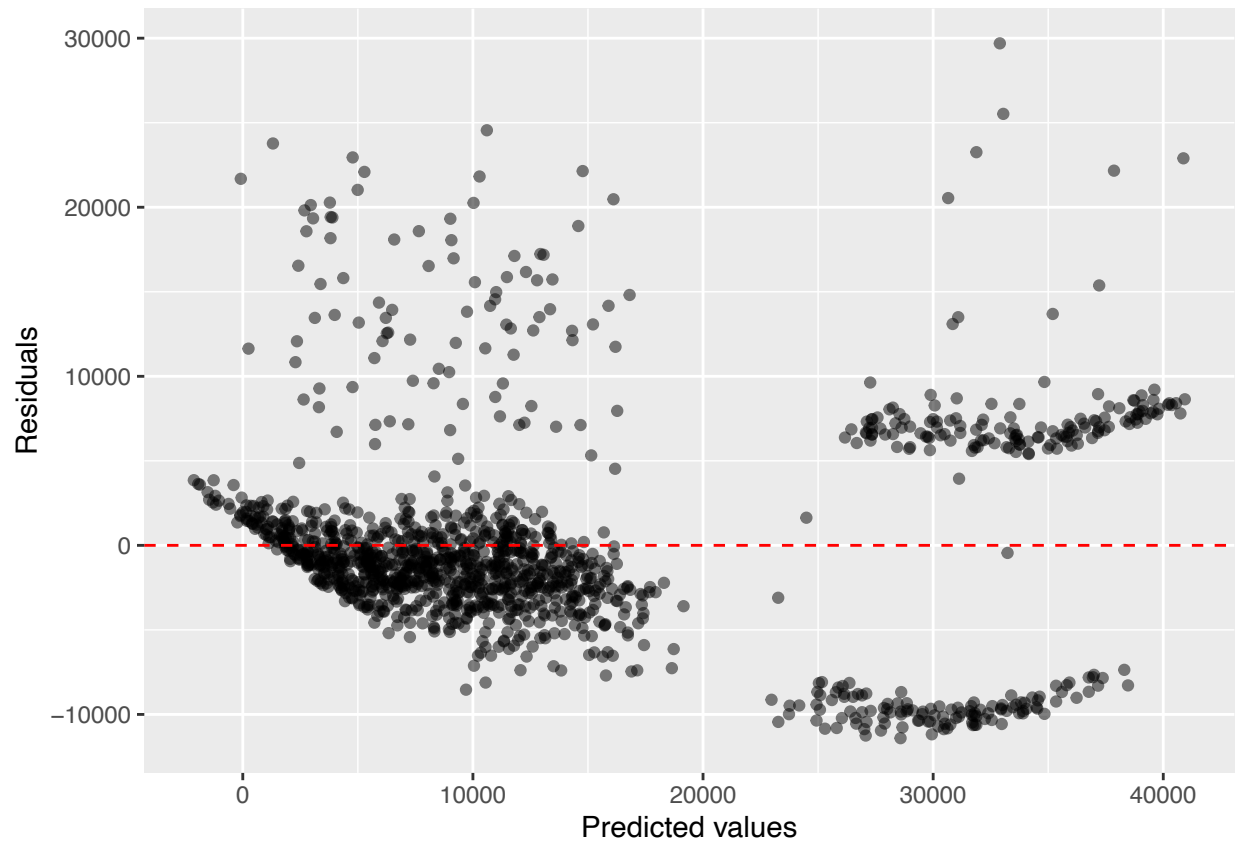
```
##
```

	Df	Sum of Sq	RSS	AIC
## + children	1	4.3477e+08	4.9078e+10	23315
## + region	1	1.8973e+08	4.9323e+10	23322

```
## <none> 4.9513e+10 23325
## + sex 1 3.9429e+06 4.9509e+10 23327
## - bmi 1 5.1129e+09 5.4626e+10 23454
## - age 1 1.7556e+10 6.7069e+10 23729
## - smoker 1 1.2358e+11 1.7310e+11 24997
##
## Step: AIC=23314.96
## charges ~ smoker + age + bmi + children
##
## Df Sum of Sq RSS AIC
## + region 1 1.9874e+08 4.8880e+10 23312
## <none> 4.9078e+10 23315
## + sex 1 5.4861e+06 4.9073e+10 23317
## - children 1 4.3477e+08 4.9513e+10 23325
## - bmi 1 5.0884e+09 5.4167e+10 23445
## - age 1 1.7297e+10 6.6375e+10 23717
## - smoker 1 1.2345e+11 1.7253e+11 24995
##
## Step: AIC=23311.53
## charges ~ smoker + age + bmi + children + region
##
## Df Sum of Sq RSS AIC
## <none> 4.8880e+10 23312
## + sex 1 5.6988e+06 4.8874e+10 23313
## - region 1 1.9874e+08 4.9078e+10 23315
## - children 1 4.4378e+08 4.9323e+10 23322
## - bmi 1 5.2802e+09 5.4160e+10 23447
## - age 1 1.7233e+10 6.6112e+10 23714
## - smoker 1 1.2341e+11 1.7229e+11 24995
```

Checking for Linearity and Constant Variance Linearity and Constant variance are not satisfied hence showing we need to continue modifying our model.

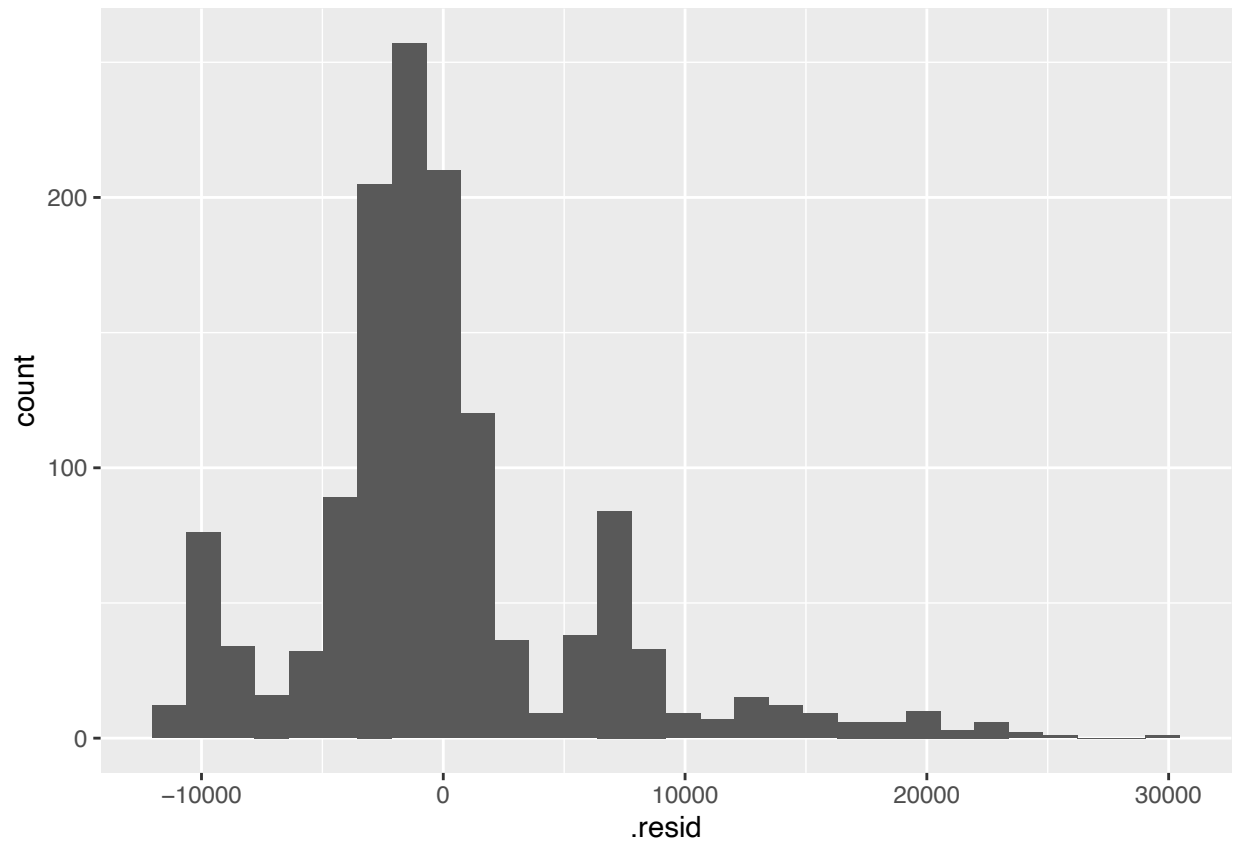
```
library(ggplot2)
ggplot(data = step_model1, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(x = "Predicted values", y = "Residuals")
```



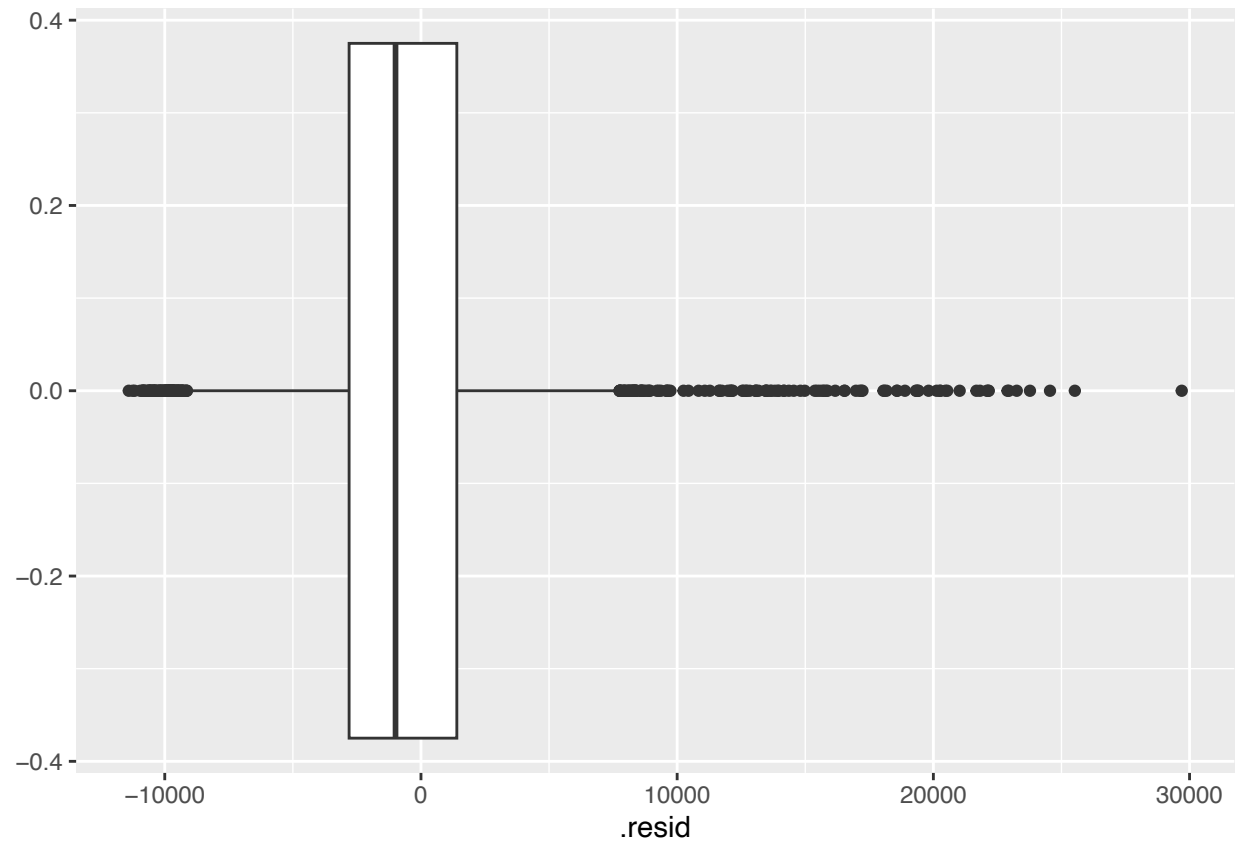
Checking for Normality The normality assumption is not satisfied as can be seen by the graphs and confirmed through the shapiro wilk test.

```
ggplot(data =step_model1, aes(x = .resid)) +geom_histogram()
```

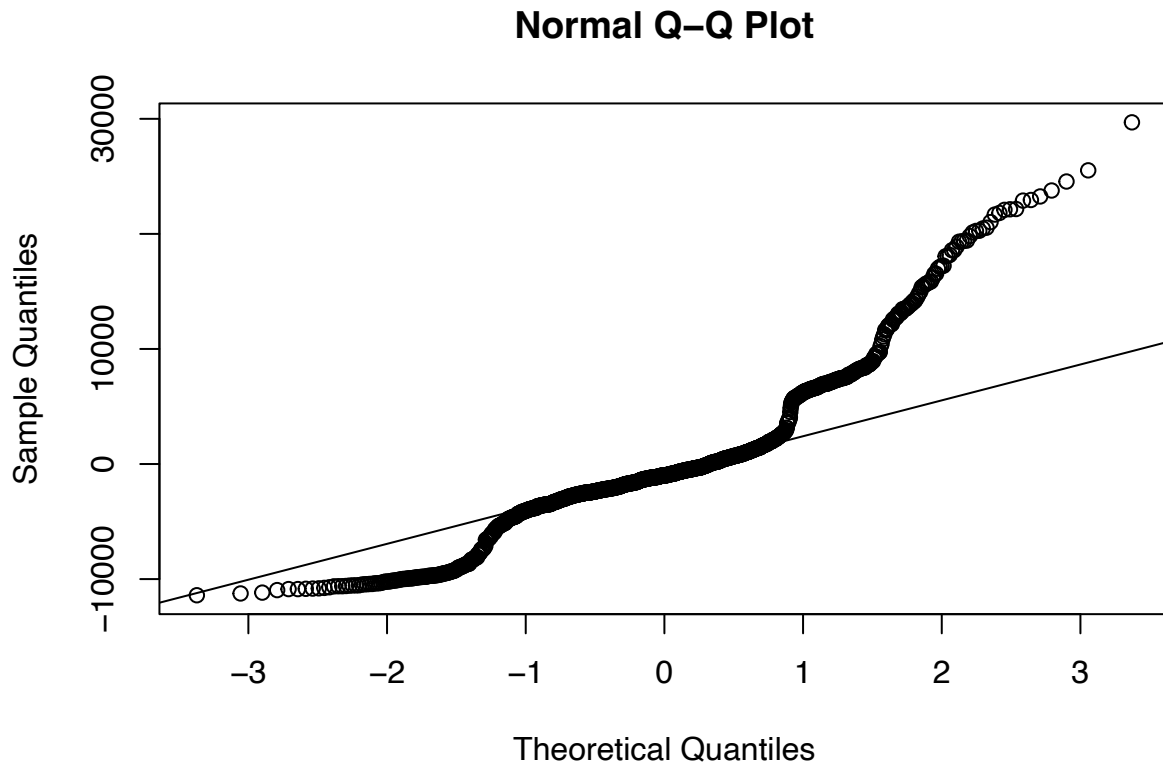
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



```
ggplot(data =step_model1, aes(x = .resid)) +geom_boxplot()
```



```
qqnorm(resid(step_model1))  
qqline(resid(step_model1))
```



Since $2.2\text{e-}16$ is less than $.05$ we reject H_0 . Hence the normality assumption is not satisfied.

```
shapiro.test(resid(step_model1))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid(step_model1)
## W = 0.89922, p-value < 2.2e-16
```

Checking for Independence Since p value is greater than 0.05 we fail to reject H_0 . Hence we can say that there is no correlation between the variables.

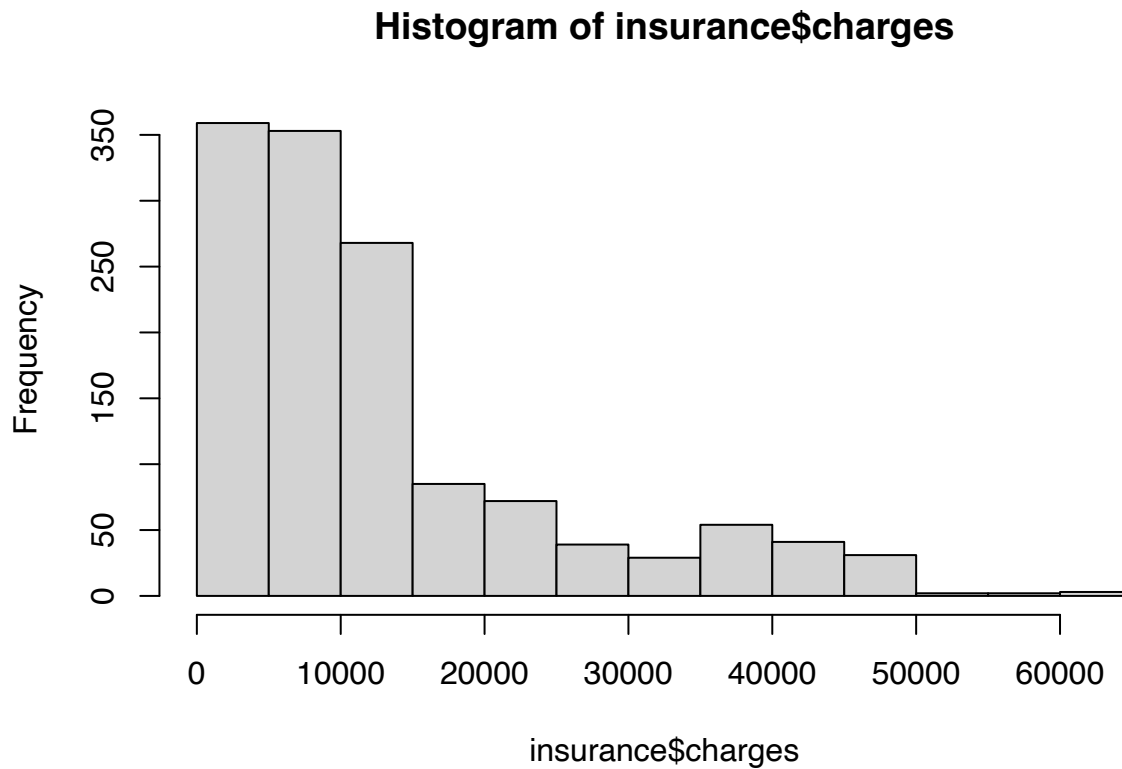
```
library(car)
dwt(step_model1)
```

```
## lag Autocorrelation D-W Statistic p-value
## 1 -0.04536255 2.088121 0.118
## Alternative hypothesis: rho != 0
```

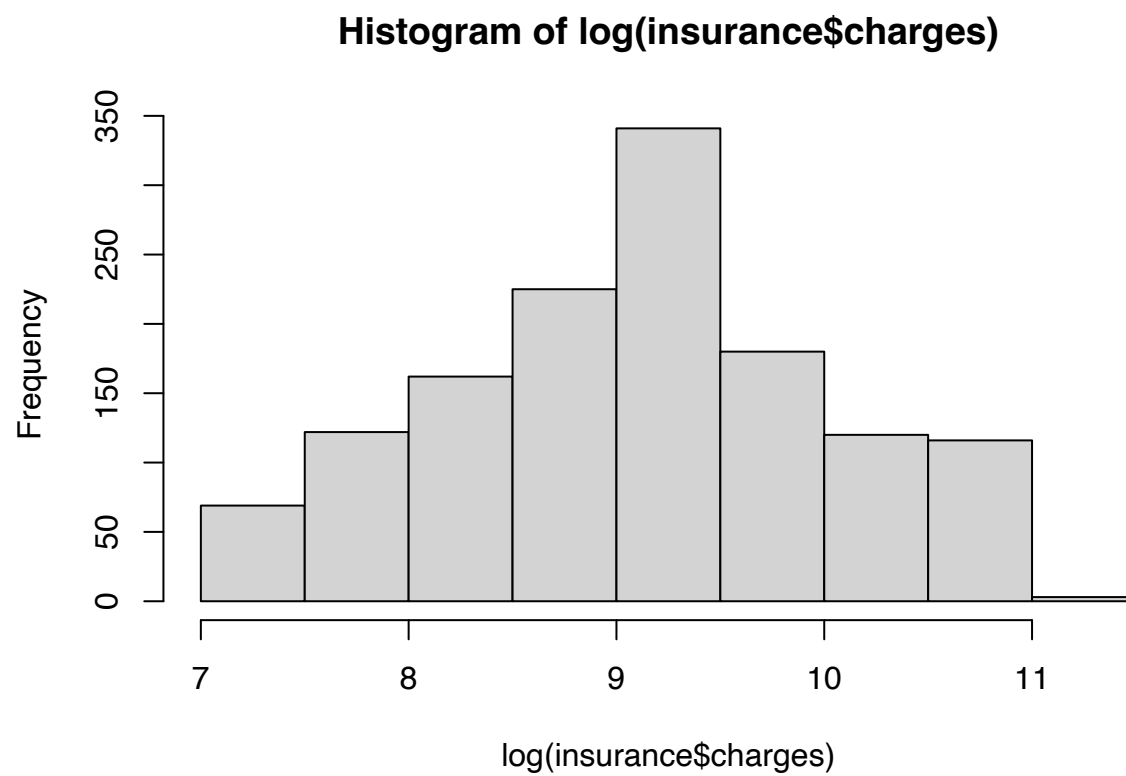
When running the model including all the predictors we can see that using stepwise regression. The sex predictor is not a very good predictor for the model since it has a p value of 0.693348 . Currently, using this model, normality linearity, and constant variance are not satisfied. Hence for our final model we may want to consider interaction in our model and make necessary transformations so the assumptions can be satisfied. Our research question was if

Male or Females pay more insurance charges. However, since we see that sex is not a good predictor we can look into another research question to look into if smokers or non-smokers pay more insurance charges depending on the region they are located.

```
hist(insurance$charges)
```

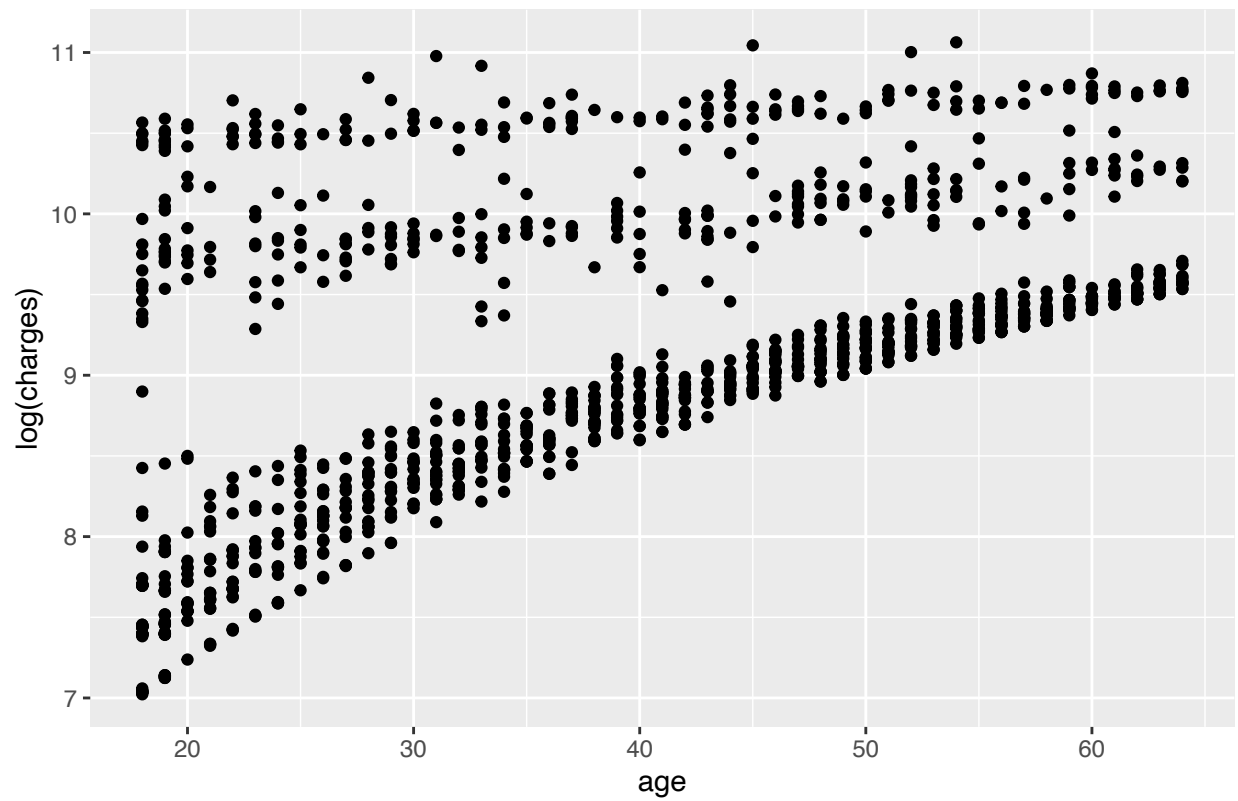


```
hist(log(insurance$charges))
```

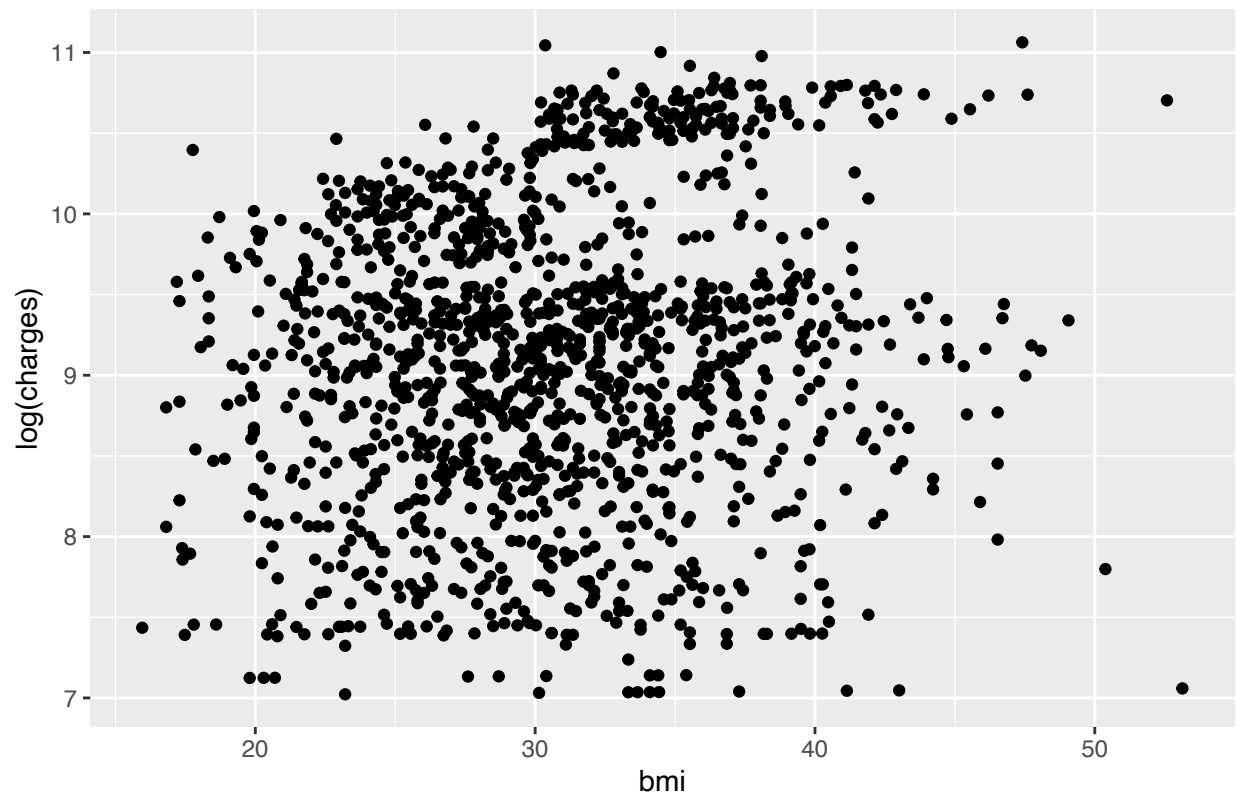
```
ggplot(insurance, aes(x = age, y = log(charges))) + geom_point() + labs(title = "Age vs Charges")
```

Age vs Charges

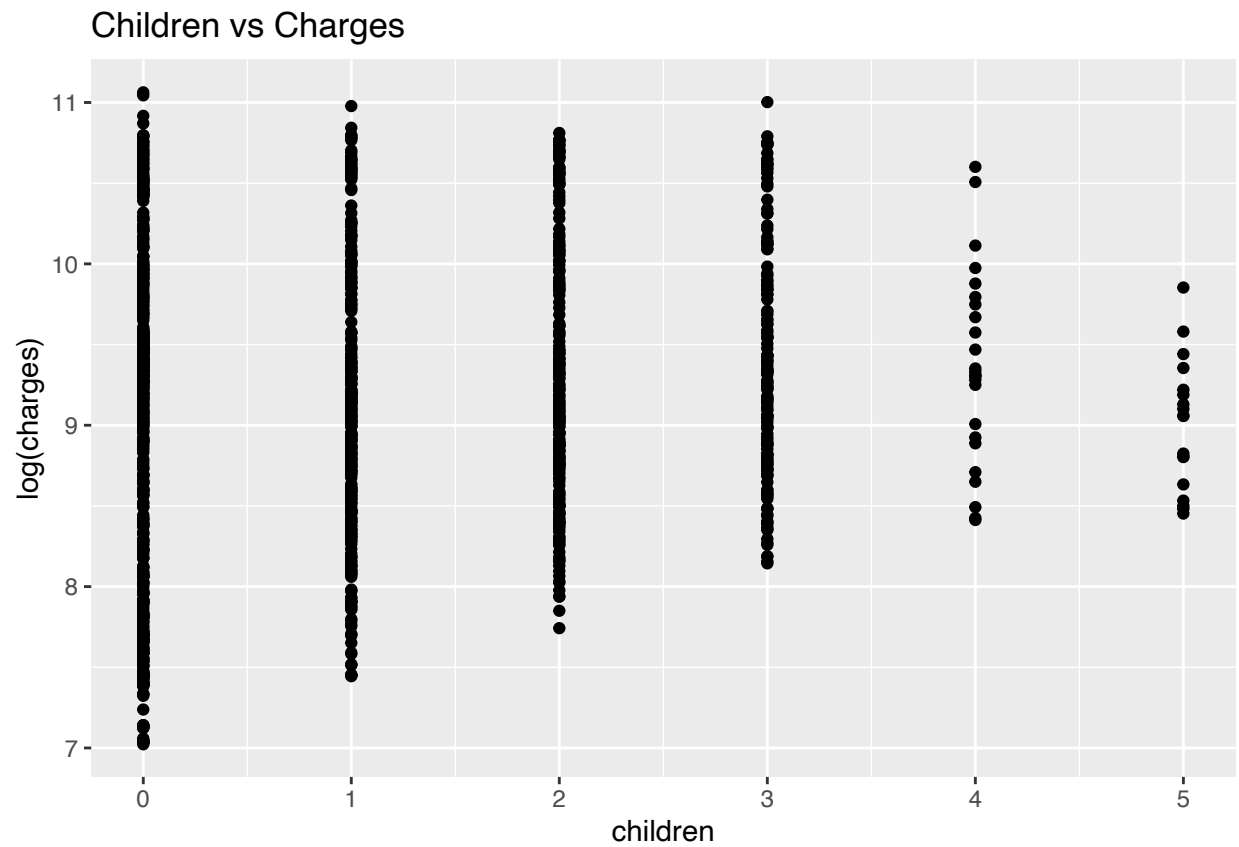


```
ggplot(insurance, aes(x = bmi, y = log(charges))) + geom_point() + labs(title = "BMI vs Charges")
```

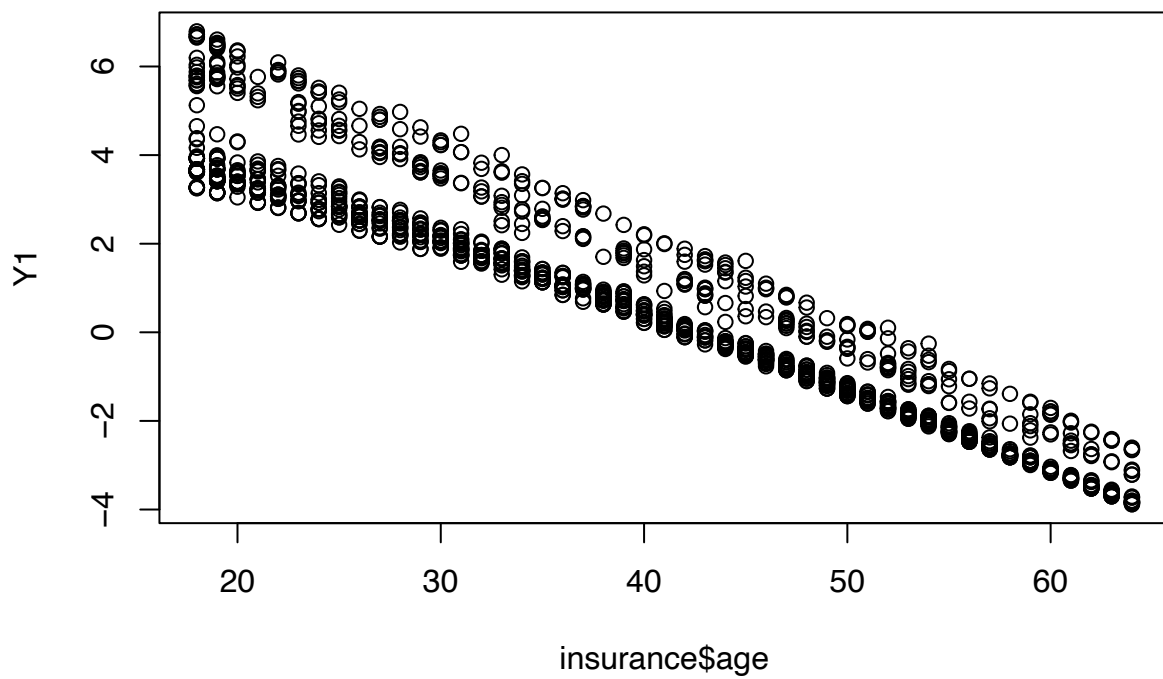
BMI vs Charges



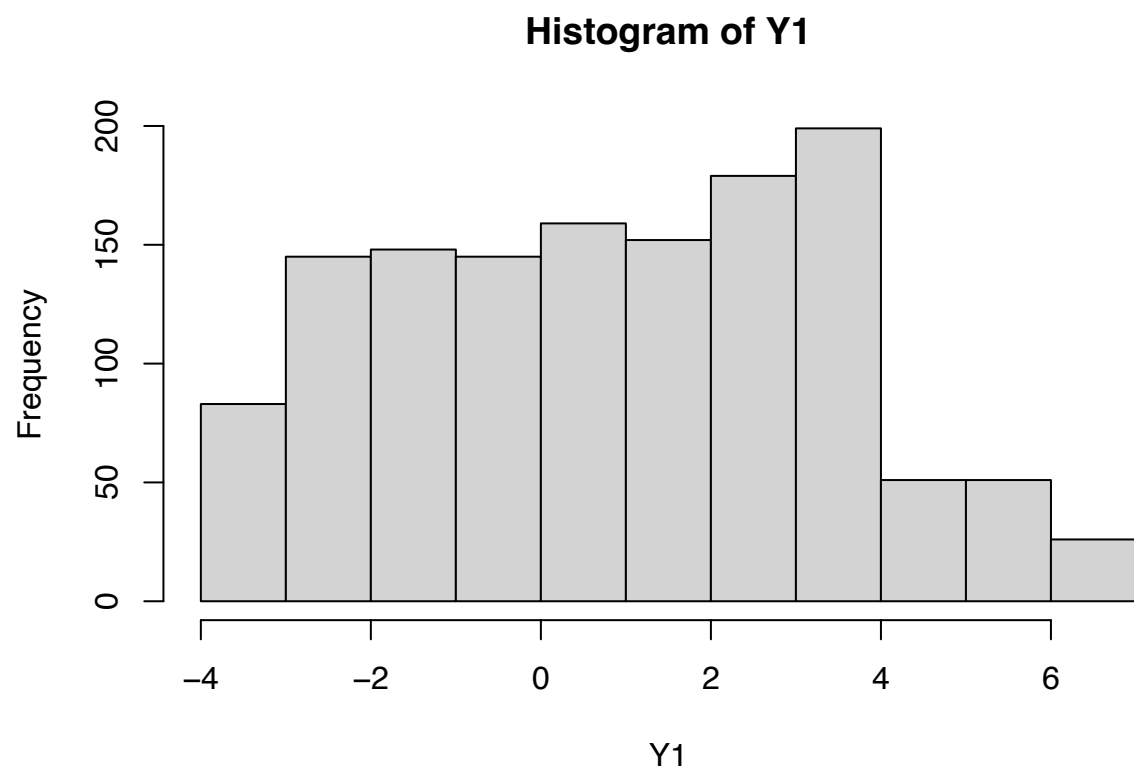
```
ggplot(insurance, aes(x = children, y =log(charges))) + geom_point() + labs(title = "Children vs Charg
```



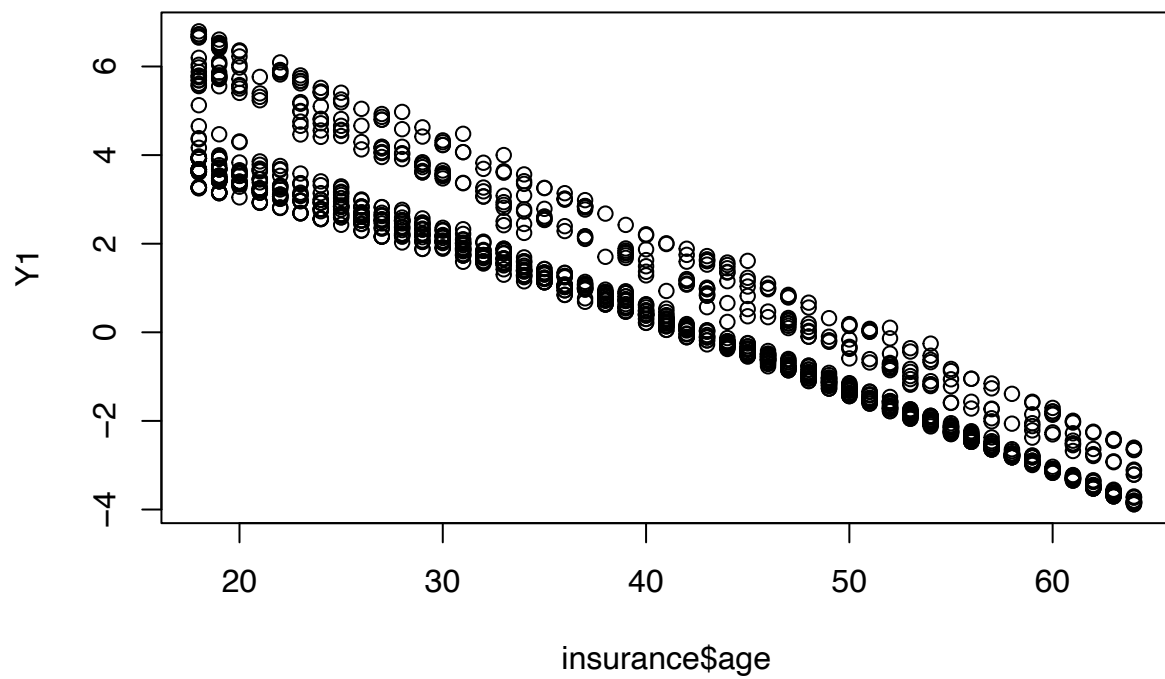
```
lm1 <- lm(log(charges) ~ 0 + age, insurance)
Y1 <- lm1$resid
plot(insurance$age, Y1)
```



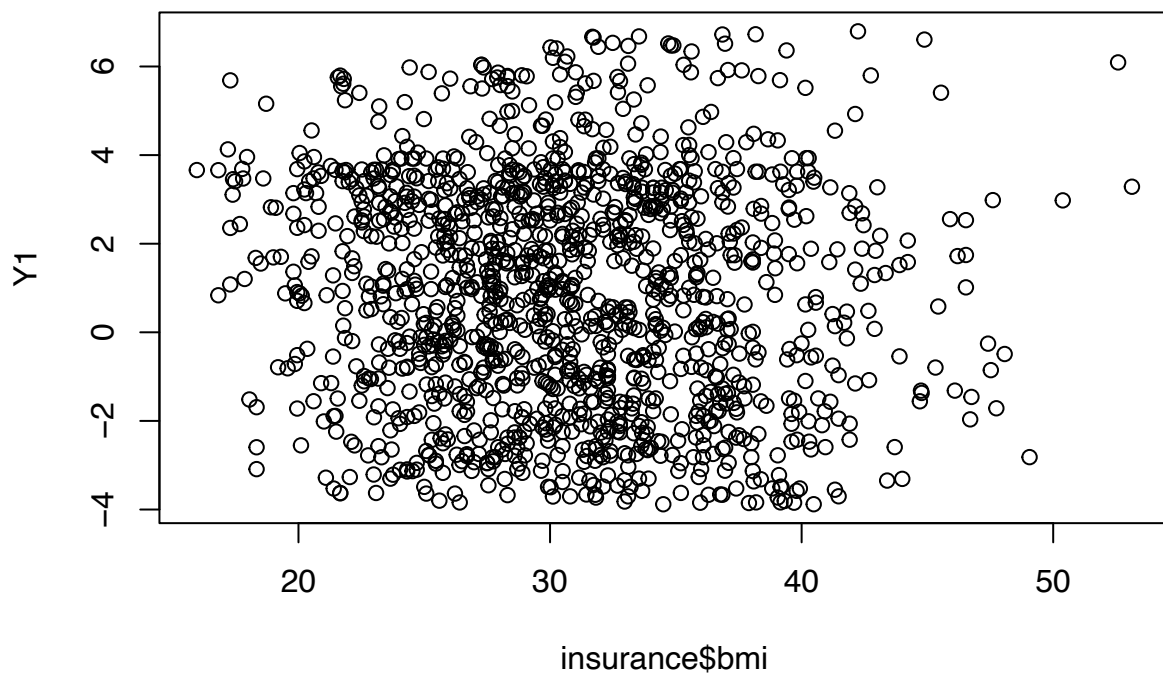
```
hist(Y1)
```



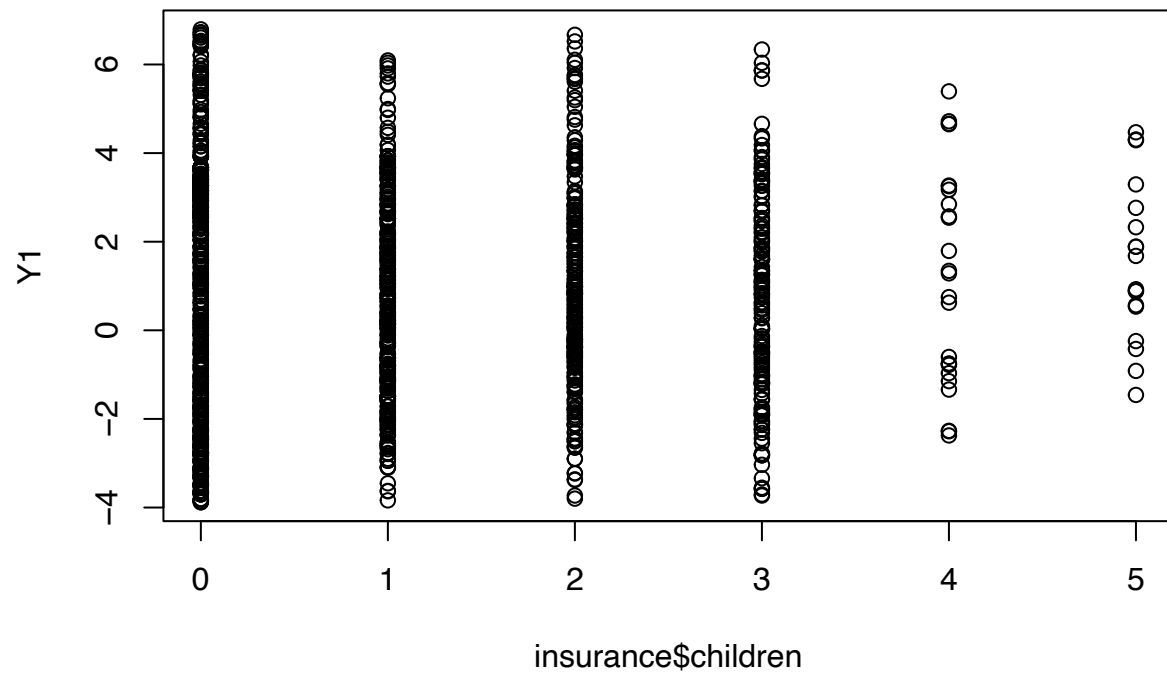
```
plot(insurance$age,Y1)
```



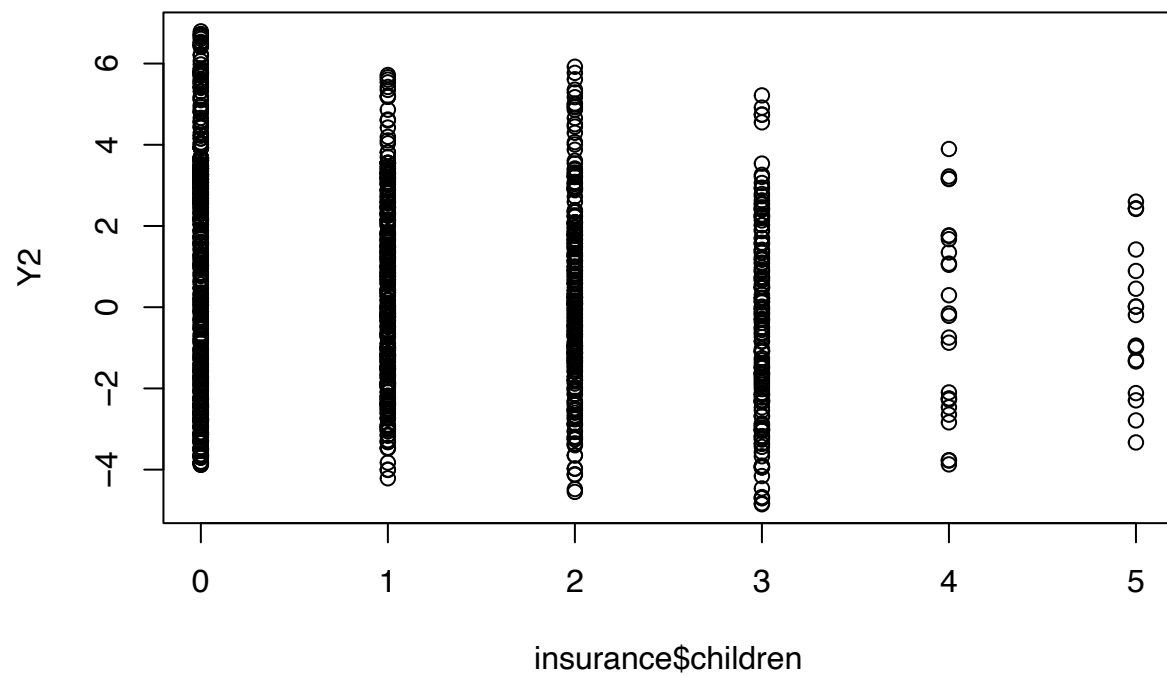
```
plot(insurance$age,Y1)
```



```
plot(insurance$children,Y1)
```

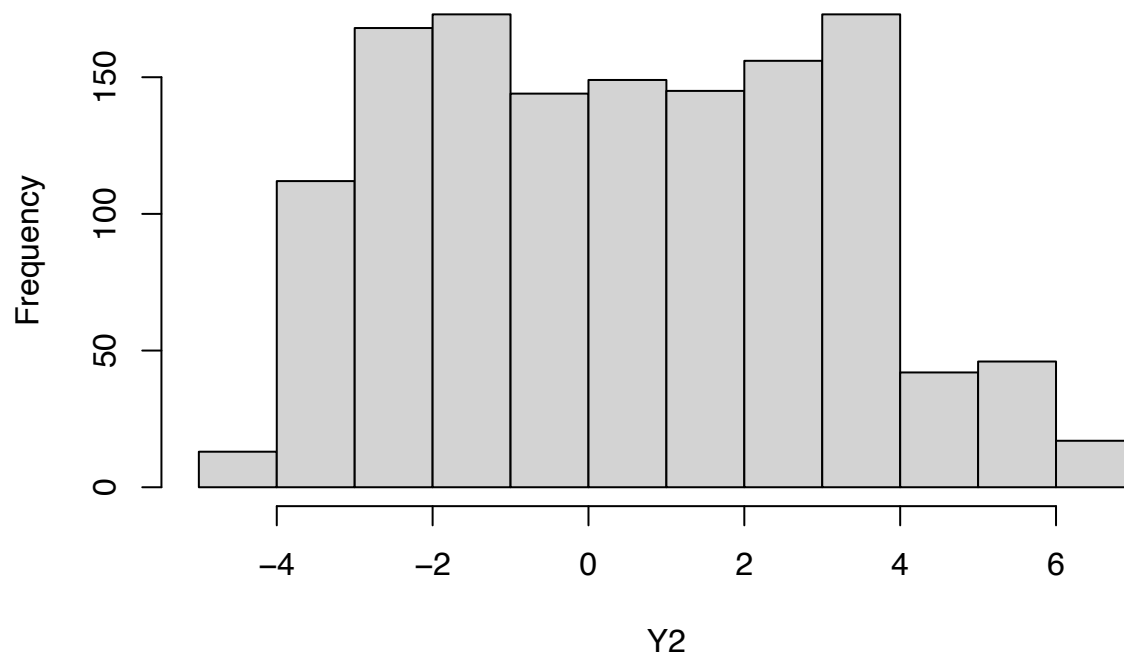



```
lm2 <- lm(Y1 ~ 0 + children, insurance)
Y2 <- lm2$residuals
plot(insurance$children, Y2)
```

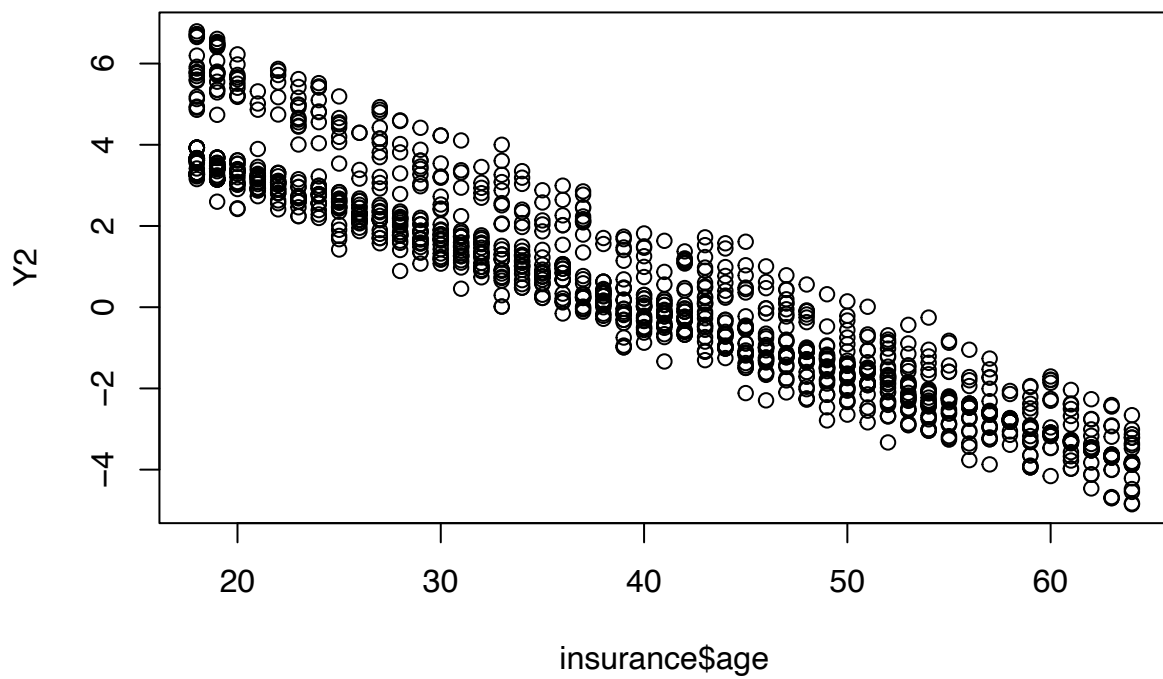


```
hist(Y2)
```

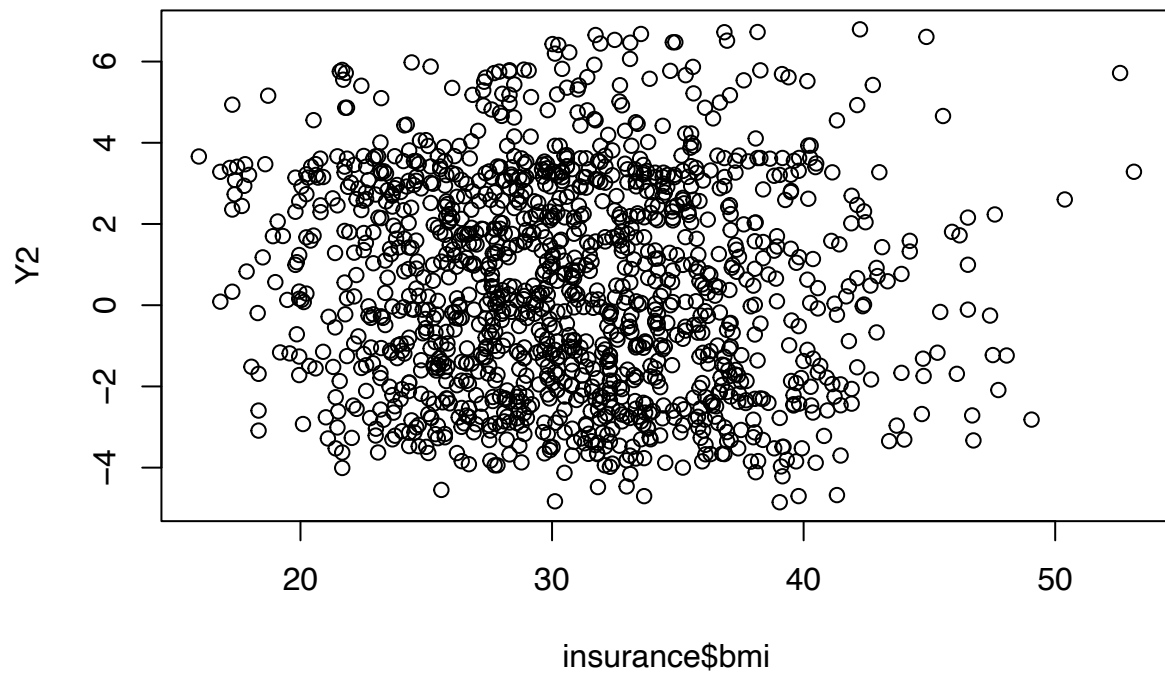
Histogram of Y2



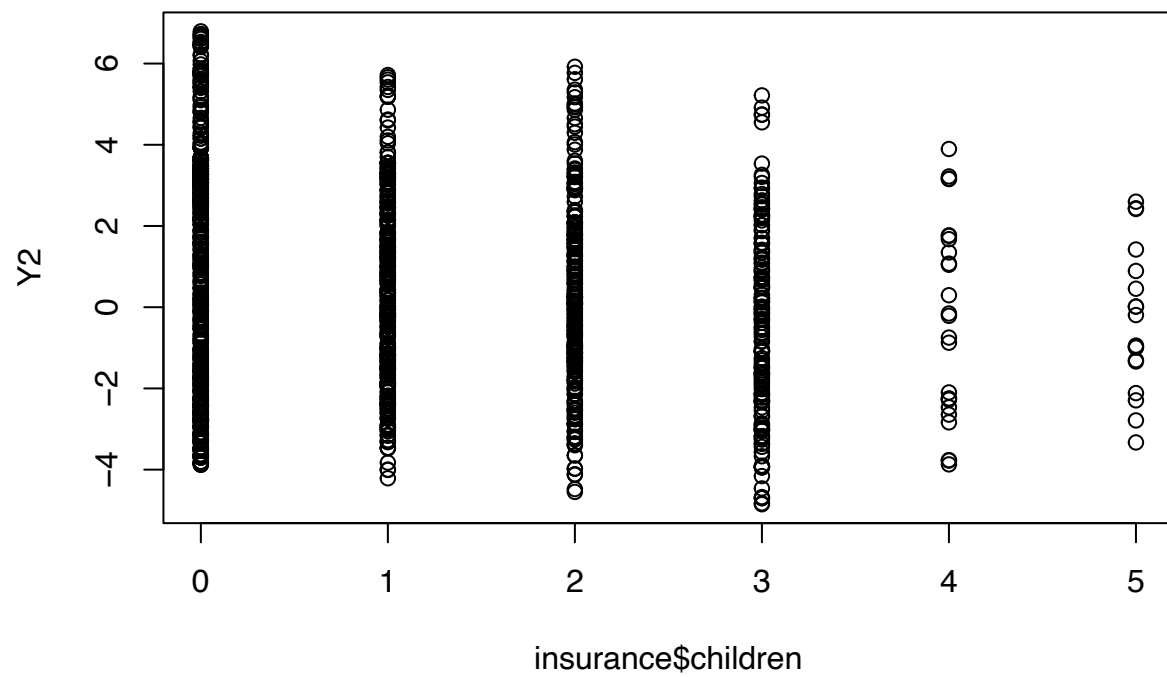
```
plot(insurance$age,Y2)
```



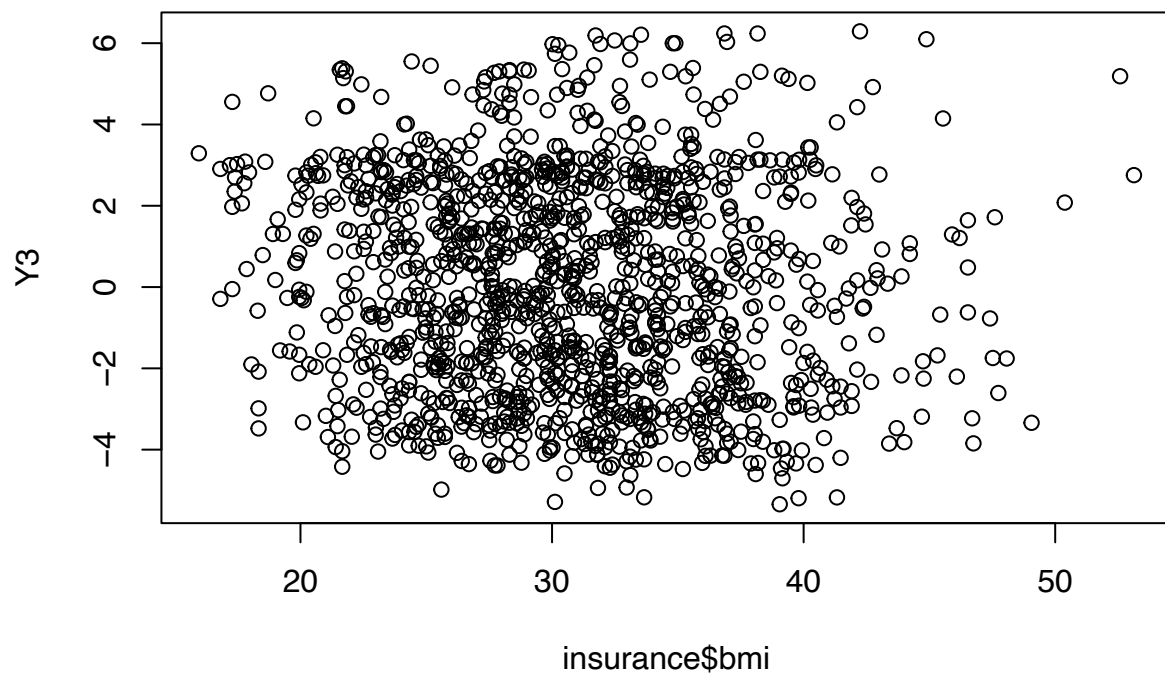
```
plot(insurance$age,Y2)
```



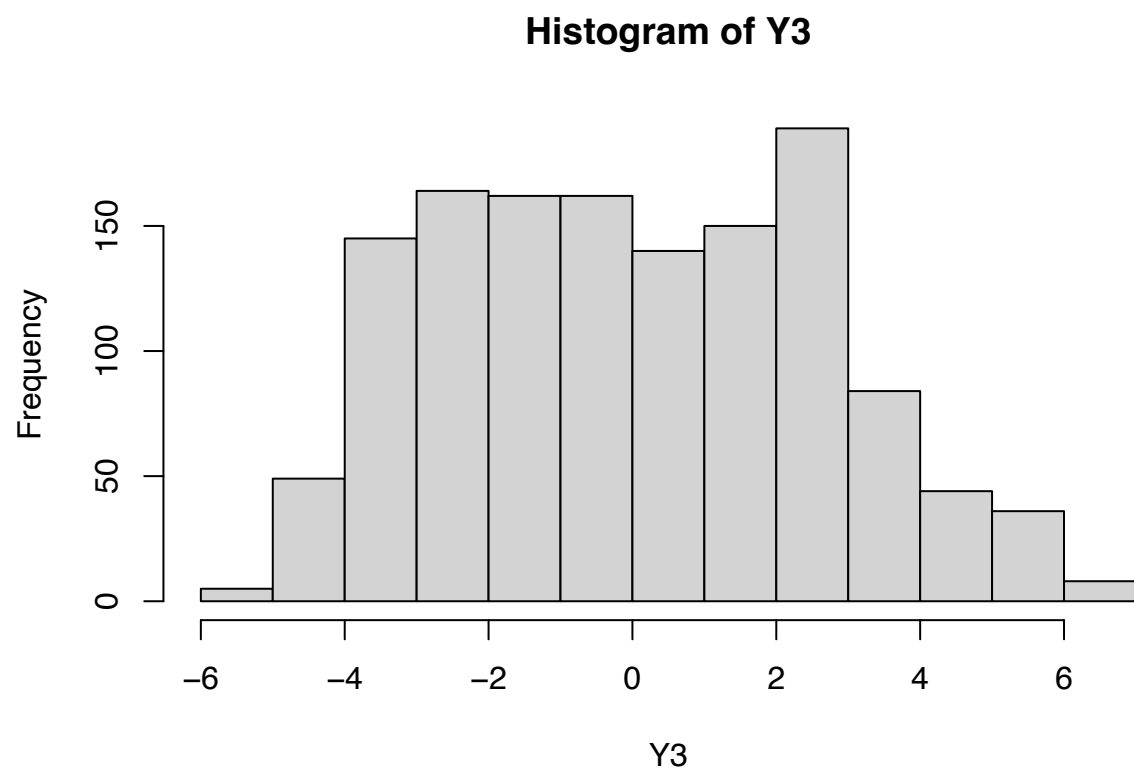
```
plot(insurance$children,Y2)
```



```
lm3 <- lm(Y2 ~ 0 + log(bmi), insurance)
Y3 <- lm3$residuals
plot(insurance$bmi, Y3)
```



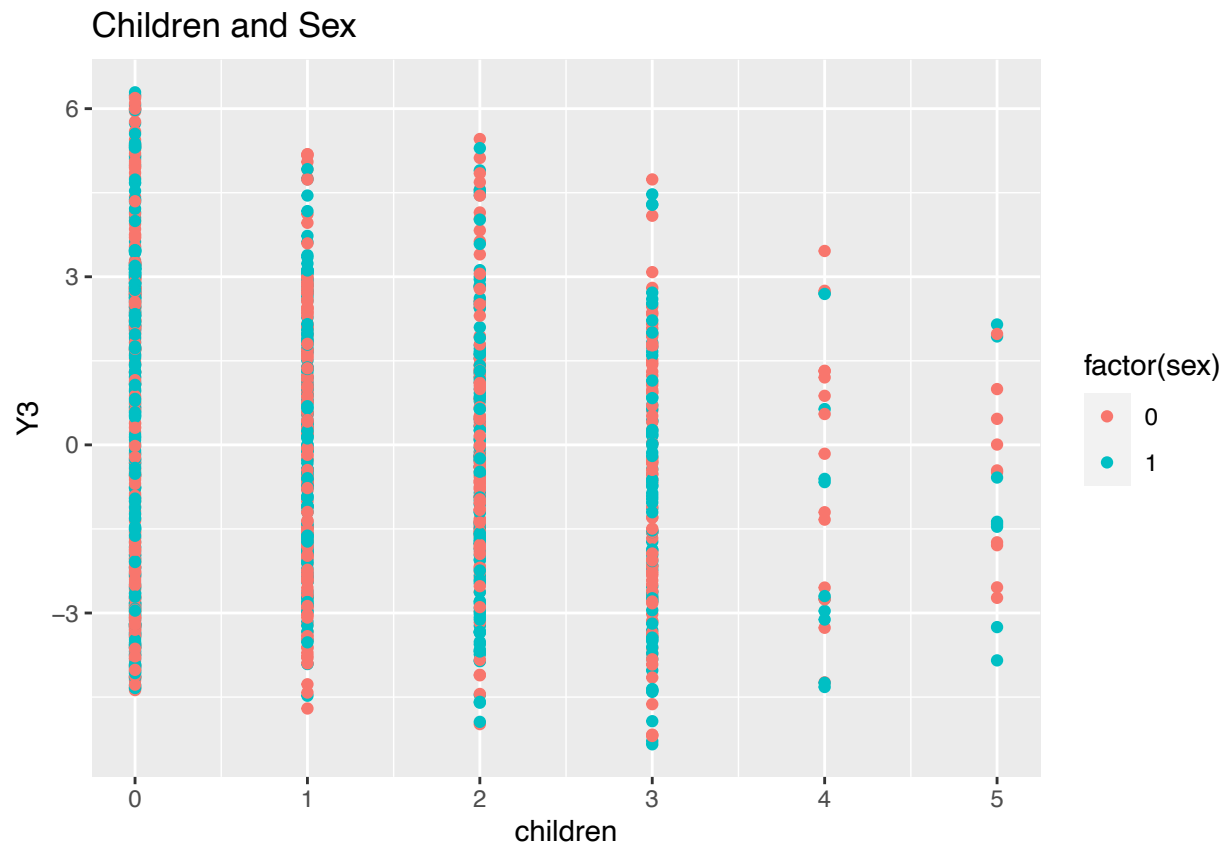
```
hist(Y3)
```



```
#interactions  
ggplot(insurance, aes(x = age, y = Y3, color = factor(sex))) +  
  geom_point() +  
  labs(title = "Age and Sex")
```

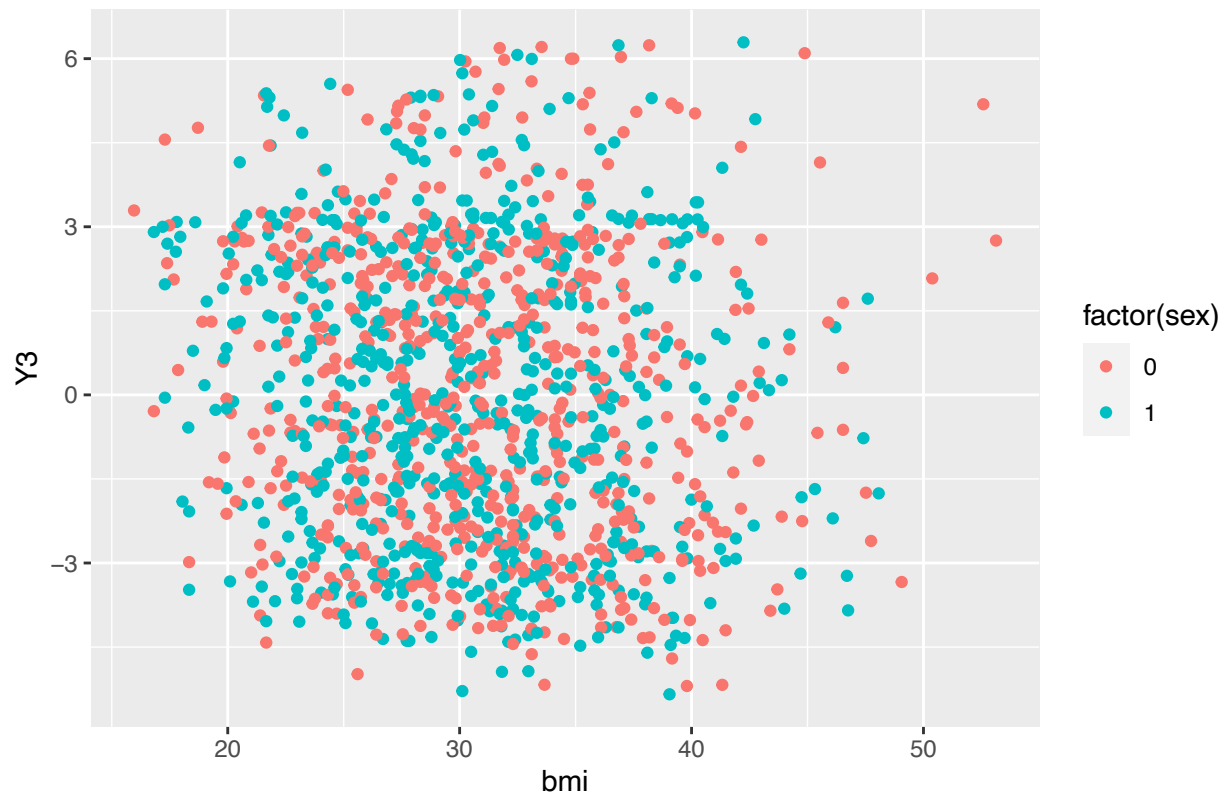



```
ggplot(insurance, aes(x = children, y = Y3, color = factor(sex))) +  
  geom_point() +  
  labs(title = "Children and Sex")
```

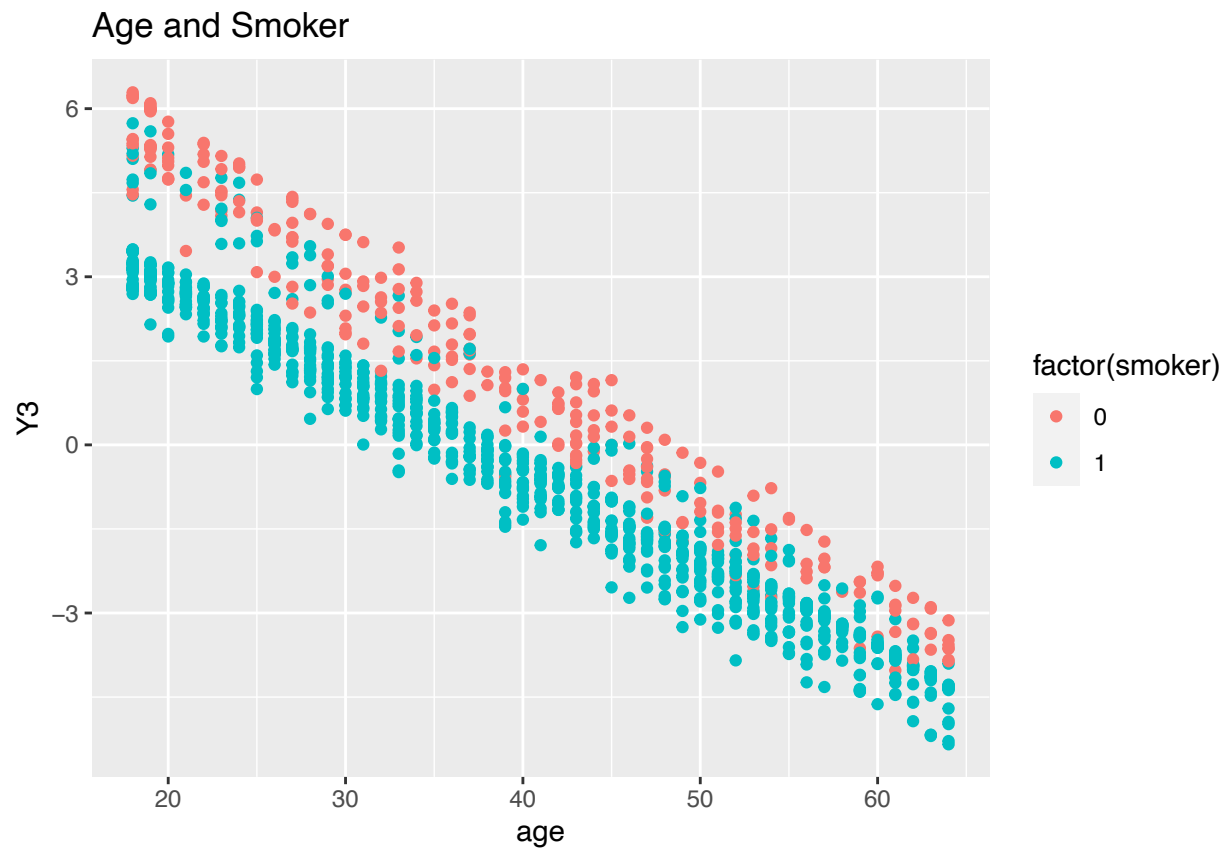


```
ggplot(insurance, aes(x = bmi, y = Y3, color = factor(sex))) +  
  geom_point() +  
  labs(title = "BMI and Sex")
```

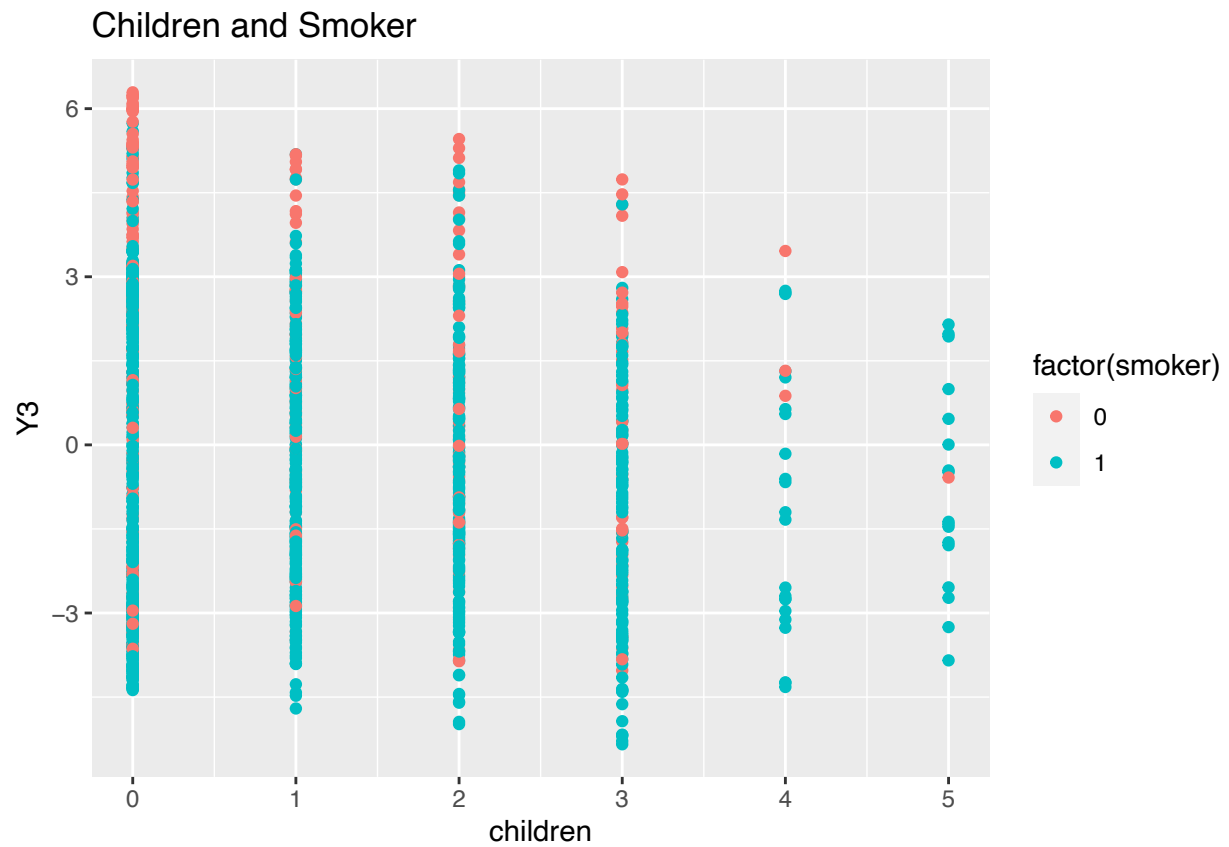
BMI and Sex



```
ggplot(insurance, aes(x = age, y = Y3, color = factor(smoker))) +  
  geom_point() +  
  labs(title = "Age and Smoker")
```

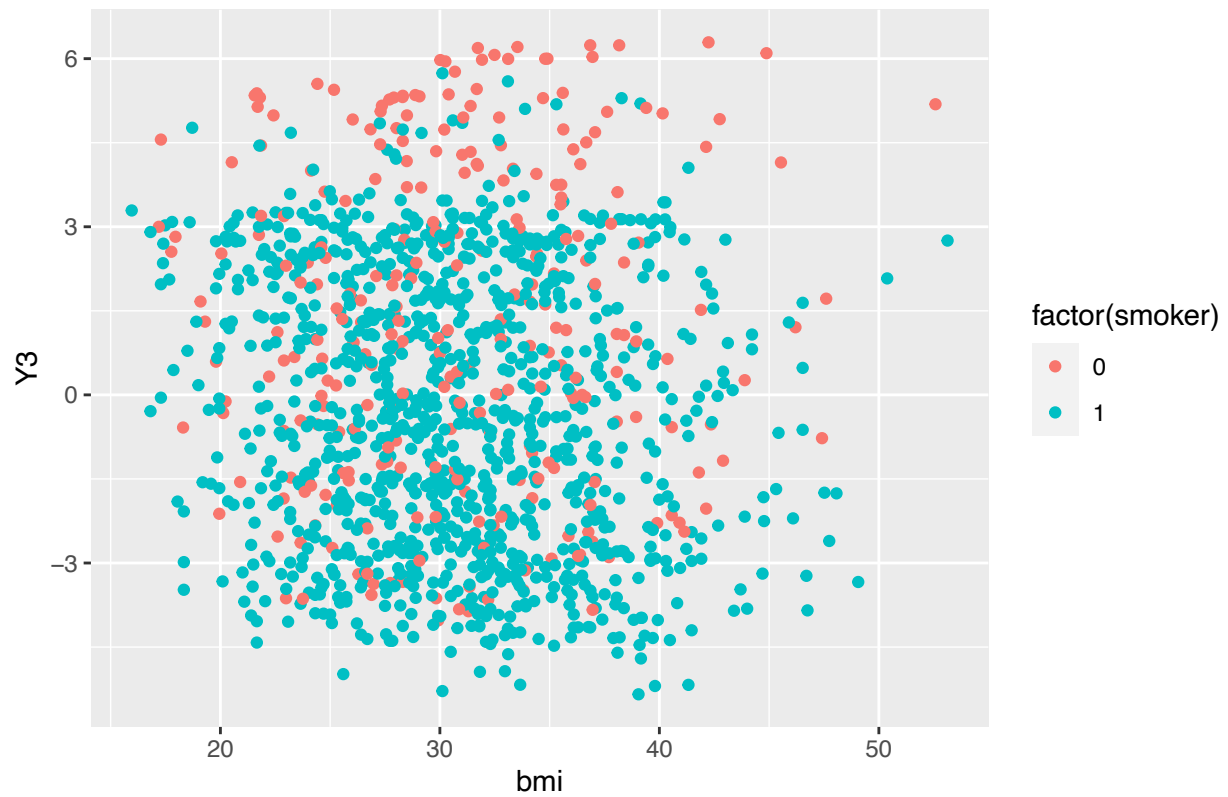


```
ggplot(insurance, aes(x = children, y = Y3, color = factor(smoker))) +  
  geom_point() +  
  labs(title = "Children and Smoker")
```

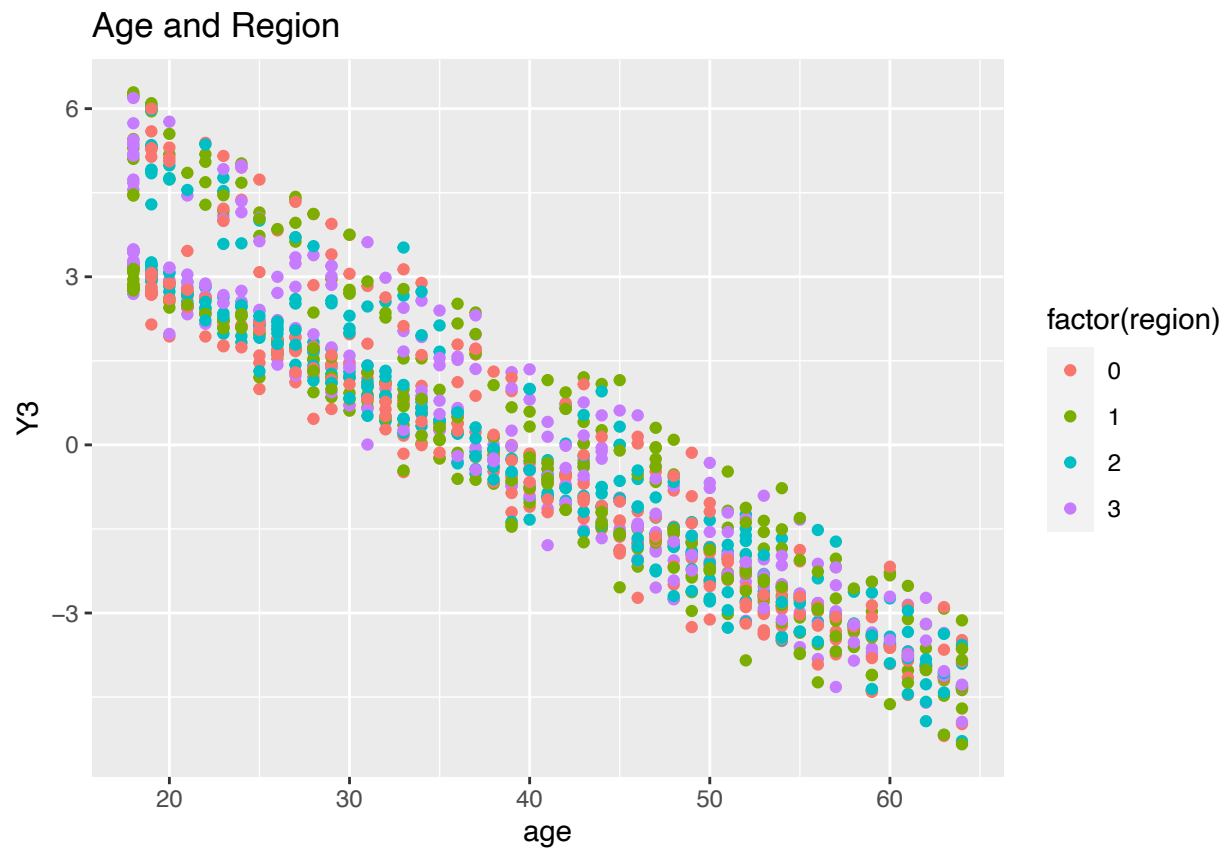


```
ggplot(insurance, aes(x = bmi, y = Y3, color = factor(smoker))) +  
  geom_point() +  
  labs(title = "BMI and Smoker")
```

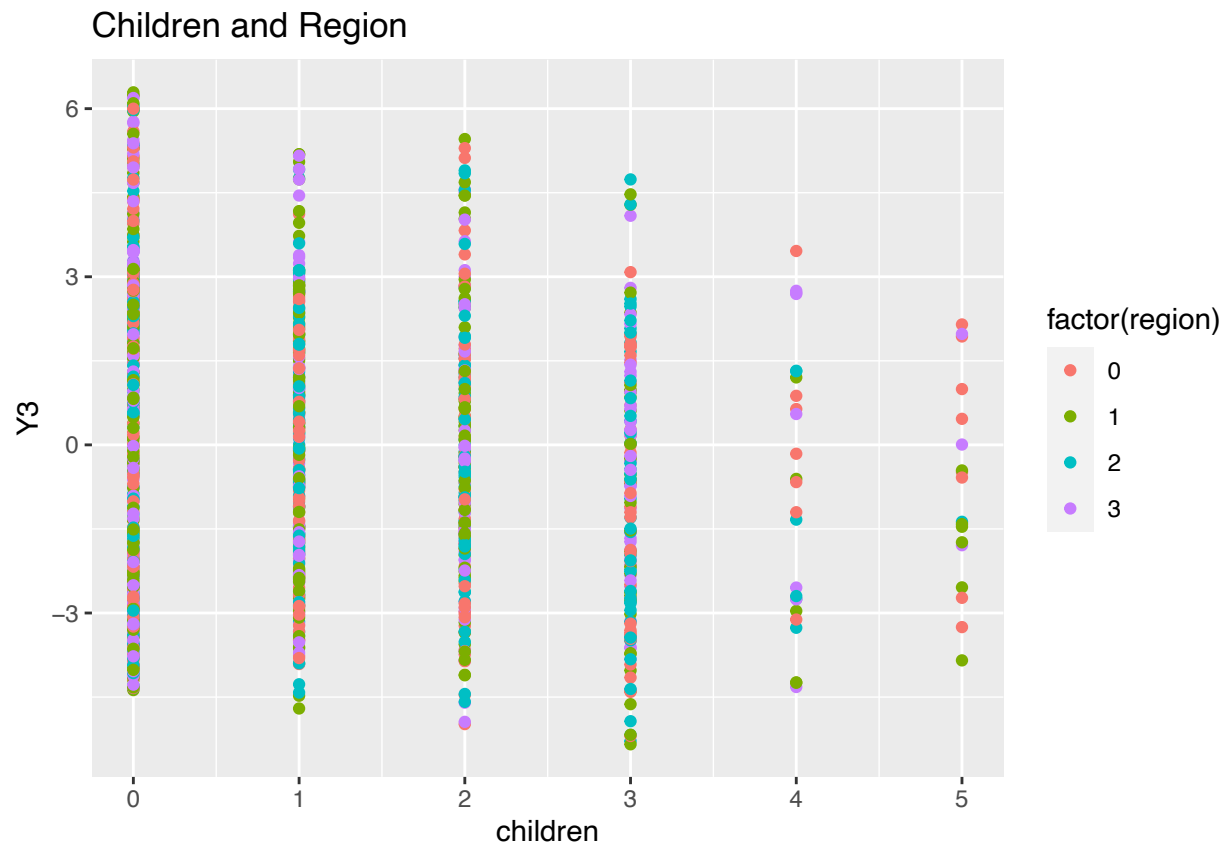
BMI and Smoker



```
ggplot(insurance, aes(x = age, y = Y3, color = factor(region))) +  
  geom_point() +  
  labs(title = "Age and Region")
```

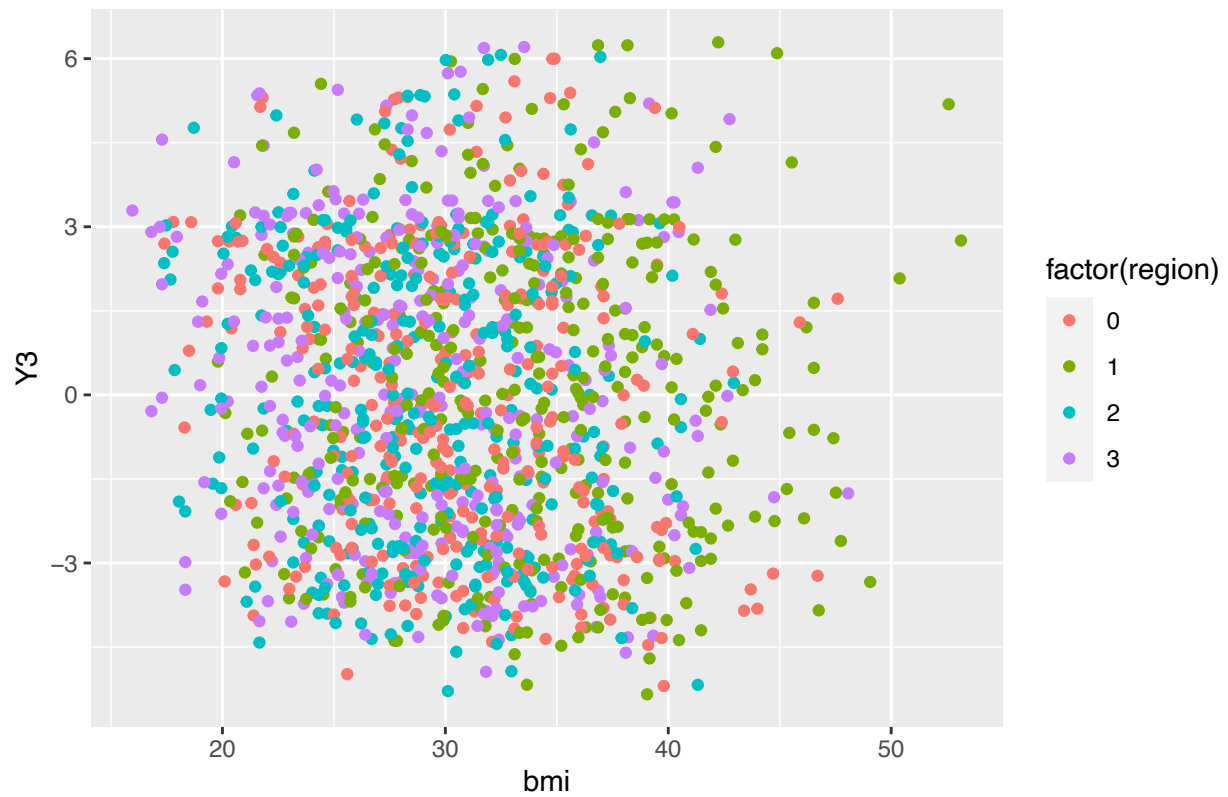


```
ggplot(insurance, aes(x = children, y = Y3, color = factor(region))) +  
  geom_point() +  
  labs(title = "Children and Region")
```

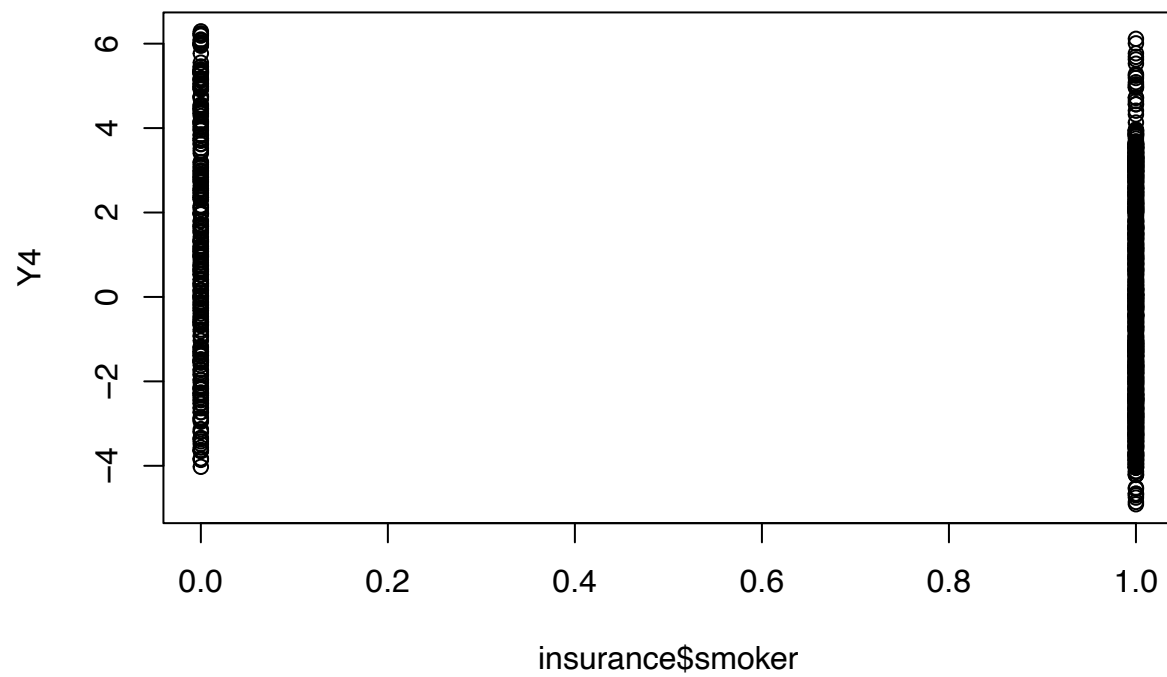


```
ggplot(insurance, aes(x = bmi, y = Y3, color = factor(region))) +  
  geom_point() +  
  labs(title = "BMI and Region")
```


BMI and Region

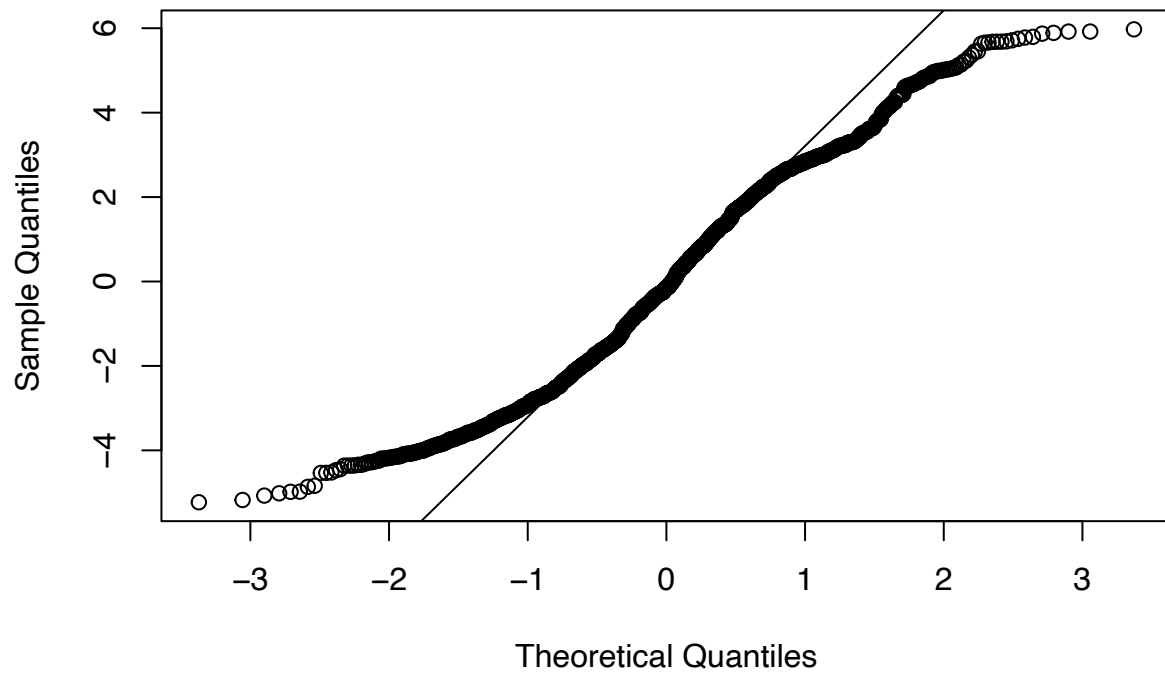


```
lm4 <- lm(Y3 ~ 0 + insurance$bmi:insurance$smoker)
Y4 <- lm4$residuals
plot(insurance$smoker, Y4)
```

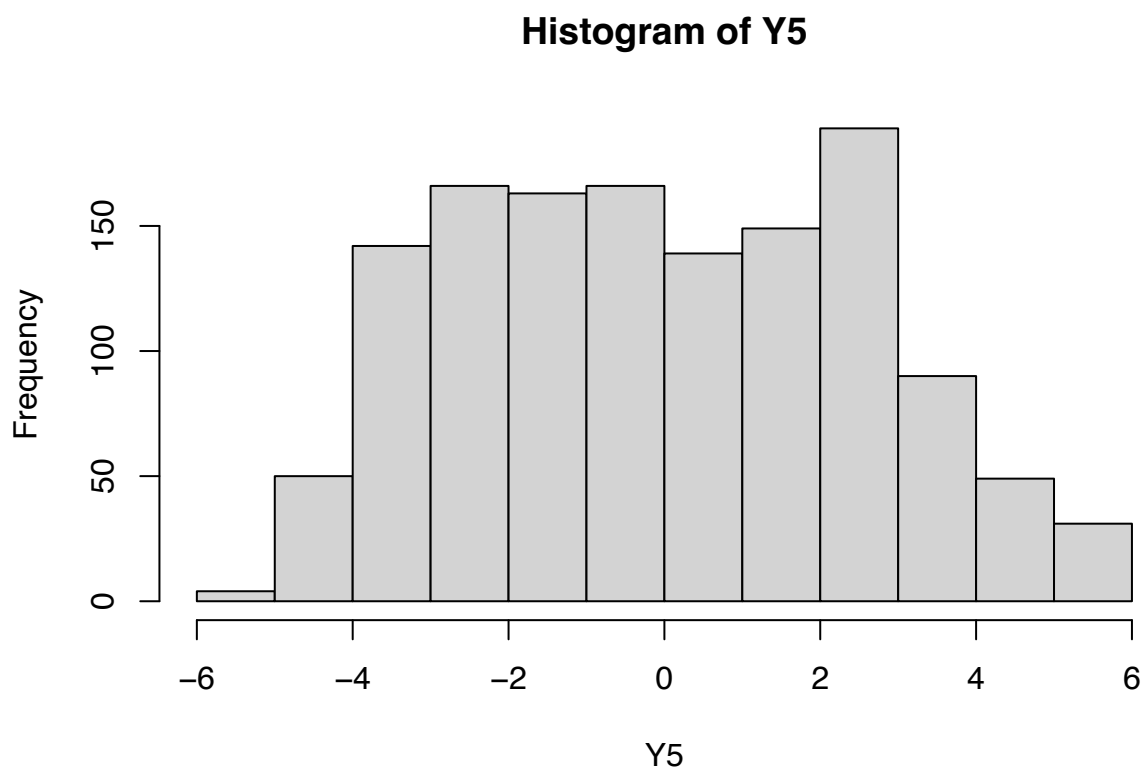


```
lm5 <- lm(Y4 ~ 1)
Y5 <- lm5$residuals
qqnorm(Y5)
qqline(Y5)
```

Normal Q-Q Plot



```
hist(Y5)
```



```
shapiro.test(Y5)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Y5
## W = 0.97287, p-value = 3.654e-15
```

Considered but excluded models

```
null_model <- lm(log(charges) ~ 1, data = insurance)

# Specify the full model using all of the potential predictors
full_model <- lm(log(charges) ~ 0 + age + children + bmi:smoker, insurance)

# Use a stepwise algorithm to build a parsimonious model
step_model1 <- step(null_model, scope = list(lower = null_model, upper = full_model), direction = "bot

## Start:  AIC=-223.51
## log(charges) ~ 1
##
##           Df Sum of Sq    RSS   AIC F value    Pr(>F)
## + bmi:smoker  1    385.32  745.15 -779.18 690.844 < 2.2e-16 ***
```

```

## + age          1      314.96  815.51 -658.46 515.977 < 2.2e-16 ***
## + children     1       29.43 1101.05 -256.79  35.705 2.941e-09 ***
## <none>                    1130.47 -223.51
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step: AIC=-779.18
## log(charges) ~ bmi:smoker
##
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## + age          1      364.77  380.38 -1676.87 1280.199 < 2.2e-16 ***
## + children     1       29.40  715.76 -831.04   54.834 2.315e-13 ***
## <none>                    745.15 -779.18
## - bmi:smoker   1      385.32 1130.47 -223.51  690.844 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step: AIC=-1676.87
## log(charges) ~ age + bmi:smoker
##
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## + children     1       21.28  359.10 -1751.90   79.056 < 2.2e-16 ***
## <none>                    380.38 -1676.87
## - age          1      364.77  745.15 -779.18 1280.199 < 2.2e-16 ***
## - bmi:smoker   1      435.13  815.51 -658.46 1527.131 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step: AIC=-1751.9
## log(charges) ~ age + children + bmi:smoker
##
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                    359.10 -1751.90
## - children     1       21.28  380.38 -1676.87   79.056 < 2.2e-16 ***
## - age          1      356.65  715.76 -831.04 1324.896 < 2.2e-16 ***
## - bmi:smoker   1      434.55  793.66 -692.81 1614.286 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
summary(step_model1)
```

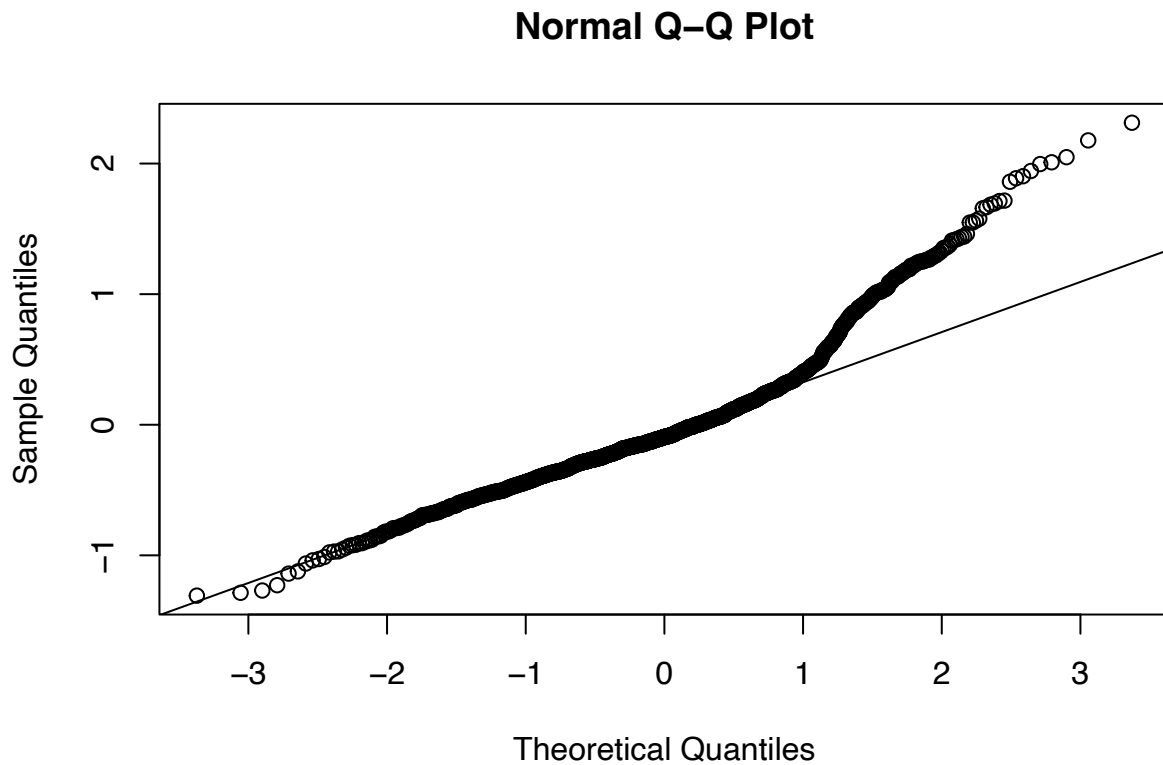
```

##
## Call:
## lm(formula = log(charges) ~ age + children + bmi:smoker, data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.30836 -0.31810 -0.09404  0.20021  2.31258
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.570103   0.049266 173.957  <2e-16 ***
## age          0.036876   0.001013  36.399  <2e-16 ***
## children     0.104752   0.011781   8.891  <2e-16 ***

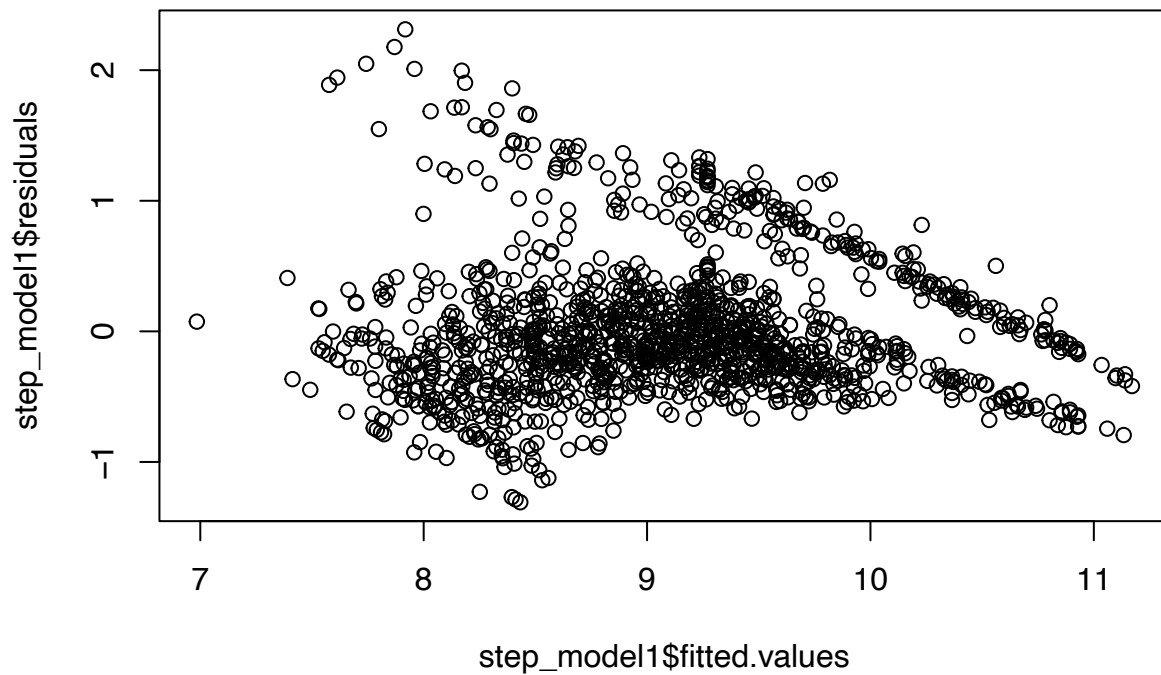
```

```
## bmi:smoker  -0.042336   0.001054 -40.178   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5188 on 1334 degrees of freedom
## Multiple R-squared:  0.6823, Adjusted R-squared:  0.6816
## F-statistic: 955.2 on 3 and 1334 DF,  p-value: < 2.2e-16
```

```
qqnorm(step_model1$residuals)
qqline(step_model1$residuals)
```



```
plot(step_model1$fitted.values, step_model1$residuals)
```



```
shapiro.test(step_model1$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  step_model1$residuals
## W = 0.92725, p-value < 2.2e-16
```

```
summary(step_model1)
```

```
##
## Call:
## lm(formula = log(charges) ~ age + children + bmi:smoker, data = insurance)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-1.30836	-0.31810	-0.09404	0.20021	2.31258

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.570103	0.049266	173.957	<2e-16 ***
age	0.036876	0.001013	36.399	<2e-16 ***
children	0.104752	0.011781	8.891	<2e-16 ***
bmi:smoker	-0.042336	0.001054	-40.178	<2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5188 on 1334 degrees of freedom
## Multiple R-squared:  0.6823, Adjusted R-squared:  0.6816
## F-statistic: 955.2 on 3 and 1334 DF,  p-value: < 2.2e-16

model1 <- lm(log(charges) ~ 0 + age + children + bmi + bmi:smoker, data = insurance)
summary(model1)
```

```
##
## Call:
## lm(formula = log(charges) ~ 0 + age + children + bmi + bmi:smoker,
##     data = insurance)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-3.8618	-0.6674	0.2620	1.1372	4.9899

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
age	0.071486	0.002464	29.014	< 2e-16 ***
children	0.245349	0.030755	7.978	3.19e-15 ***
bmi	0.229045	0.004036	56.753	< 2e-16 ***
bmi:smoker	-0.050251	0.002957	-16.996	< 2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.366 on 1334 degrees of freedom
## Multiple R-squared:  0.9777, Adjusted R-squared:  0.9777
## F-statistic: 1.465e+04 on 4 and 1334 DF,  p-value: < 2.2e-16
```

```
model2 <- lm(log(charges) ~ 0 + children + I(age^2) + I(children^2) + smoker +
  bmi:smoker + I(bmi^2):smoker, data = insurance)
summary(model2)
```

```
##
## Call:
## lm(formula = log(charges) ~ 0 + children + I(age^2) + I(children^2) +
##     smoker + bmi:smoker + I(bmi^2):smoker, data = insurance)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-5.5673	-1.1241	0.4785	2.7909	9.9414

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
children	2.525e+00	1.829e-01	13.805	< 2e-16 ***
I(age^2)	1.928e-03	6.888e-05	27.985	< 2e-16 ***
I(children^2)	-4.637e-01	5.252e-02	-8.830	< 2e-16 ***
smoker	6.444e+00	2.001e+00	3.221	0.00131 **
smoker:bmi	-1.286e-01	1.287e-01	-0.999	0.31789
smoker:I(bmi^2)	1.434e-03	2.025e-03	0.708	0.47914


```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.285 on 1332 degrees of freedom
## Multiple R-squared:  0.8716, Adjusted R-squared:  0.871
## F-statistic: 1506 on 6 and 1332 DF,  p-value: < 2.2e-16

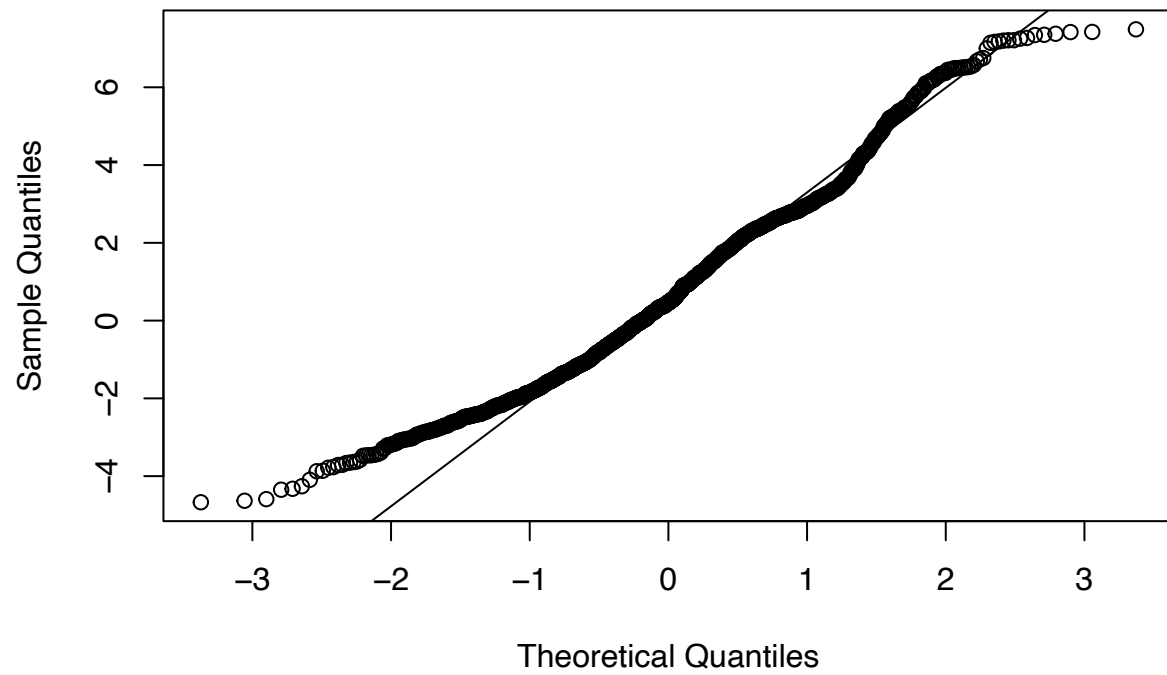
# use stepwise algorithm
step_model <- lm(formula = log(charges) ~ insurance$age + insurance$children +
  insurance$bmi + insurance$bmi:insurance$smoker, data = insurance)
summary(step_model)

##
## Call:
## lm(formula = log(charges) ~ insurance$age + insurance$children +
##     insurance$bmi + insurance$bmi:insurance$smoker, data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.90942 -0.17216 -0.04302  0.04498  2.13049
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.3115395   0.0672676   108.69  <2e-16 ***
## insurance$age      0.0350247   0.0008535    41.04  <2e-16 ***
## insurance$children  0.1024554   0.0098844    10.37  <2e-16 ***
## insurance$bmi      0.0496172   0.0020923    23.71  <2e-16 ***
## insurance$bmi:insurance$smoker -0.0500397  0.0009418   -53.13  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4353 on 1333 degrees of freedom
## Multiple R-squared:  0.7766, Adjusted R-squared:  0.7759
## F-statistic: 1158 on 4 and 1333 DF,  p-value: < 2.2e-16
```

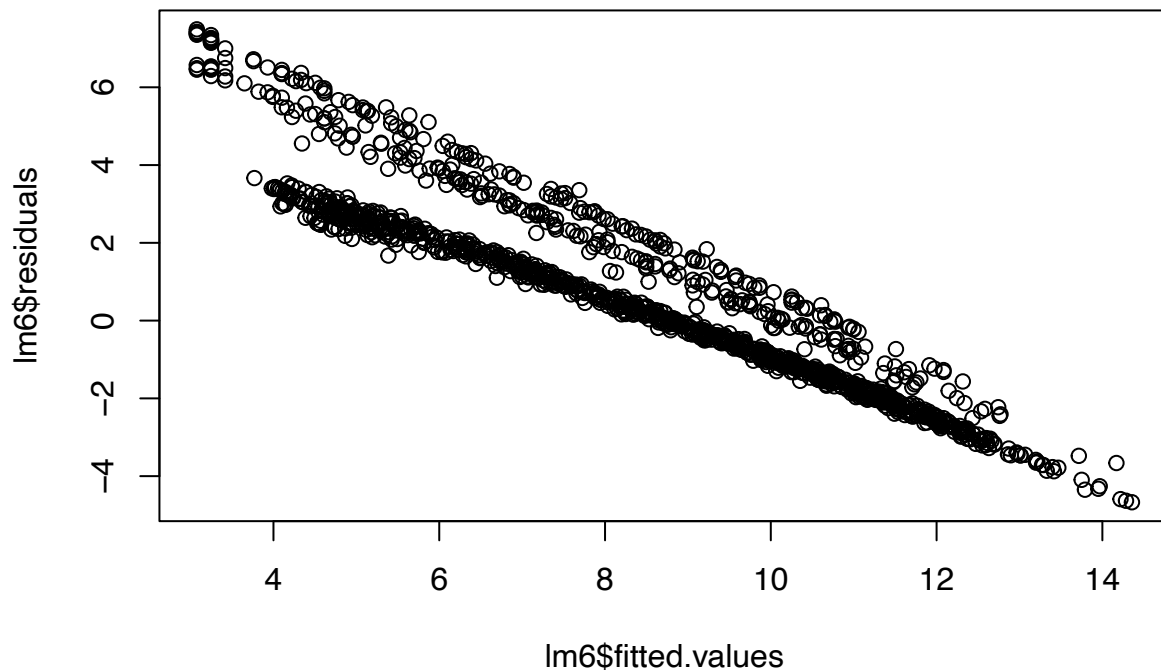
Final Model/Model Diagnostics

```
#final model
lm6 <- lm(log(charges) ~ 0 + age + children + bmi:smoker, insurance)
qqnorm(lm6$residuals)
qqline(lm6$residuals)
```

Normal Q-Q Plot



```
plot(lm6$fitted.values, lm6$residuals)
```



```
shapiro.test(lm6$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  lm6$residuals
## W = 0.97865, p-value = 3.498e-13
```

```
summary(lm6)
```

```
##
## Call:
## lm(formula = log(charges) ~ 0 + age + children + bmi:smoker,
##     data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6712 -1.2086  0.4616  2.4206  7.4904
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## age             0.170868   0.003201  53.372  <2e-16 ***
## children        0.574201   0.055791  10.292  <2e-16 ***
## bmi:smoker      0.043500   0.004529   9.604  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.524 on 1335 degrees of freedom
## Multiple R-squared:  0.924, Adjusted R-squared:  0.9238
## F-statistic: 5410 on 3 and 1335 DF, p-value: < 2.2e-16
```

When trying to find the functional form of all of our predictors in our model we found that one interaction term was needed. This interaction was between bmi and smoker. The reason this interaction term was added to our final model was because of the results we found from the bmi vs charges graph. In this graph we set non-smokers and smokers to their own color. This showed a cluster of non-smokers and smokers. This gave insight that there is an interaction between bmi and smoker.

Model Transformations

We tried various different transformations to ensure that the model assumptions were met. We first tried sqrt, log, BOXCOX, cube, and reciprocal on the x values to see if we can find linearity. We ended up not making any transformations to x since it was messing up with the normality. We transformed charges using log to see normality. This was successful so we used log of charges in our final model. Independence was also valid in our model. Constant variance unfortunately was not valid. The transformations to x and y did not help constant variance.

Multicollinearity analysis - Final Model

When running VIF we found that the values were all under 10. We removed the BMI factor since it was not outputting a VIF value since the addition of the BMI was causing there me perfect multicollinearity in our model.

```
lm6<-lm(log(insurance$charges)~0+insurance$age+insurance$children+(insurance$bmi):insurance$smoker)
summary(lm6)
```

```
##
## Call:
## lm(formula = log(insurance$charges) ~ 0 + insurance$age + insurance$children +
##      (insurance$bmi):insurance$smoker)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6712 -1.2086  0.4616  2.4206  7.4904
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## insurance$age                0.170868   0.003201  53.372   <2e-16 ***
## insurance$children            0.574201   0.055791  10.292   <2e-16 ***
## insurance$bmi:insurance$smoker 0.043500   0.004529   9.604   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.524 on 1335 degrees of freedom
## Multiple R-squared:  0.924, Adjusted R-squared:  0.9238
## F-statistic: 5410 on 3 and 1335 DF, p-value: < 2.2e-16
```

The F-test for β params = 0 is highly significant ($F=1.465e+04$, $p\text{-value} < 2.23e-16$). Therefore we can reject H_0 for any α greater than 0.0001 and conclude that at least one of the β parameters is nonzero.

```
library(car)
vif(lm6)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
## Warning in vif.default(lm6): No intercept: vifs may not be sensible.
```

```
##               insurance$age               insurance$children
##               3.733477               1.732934
## insurance$bmi:insurance$smoker
##               3.344027
```